```go
/*
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*/

package pdfcpu

import (
    "bufio"
    "bytes"
    "io"
    "os"
    "sort"
    "strconv"
    "strings"

    "github.com/pdfcpu/pdfcpu/pkg/filter"
    "github.com/pdfcpu/pdfcpu/pkg/log"
    "github.com/pkg/errors"
)

const (
    defaultBufSize = 1024
    //unknownDelimiter = byte(0)
)

// ReadFile reads in a PDF file and builds an internal structure holding its cross reference table aka the Context.
func ReadFile(inFile string, conf *Configuration) (*Context, error) {

    log.Info.Printf("reading %s..\n", inFile)

    f, err := os.Open(inFile)
    if err != nil {
        return nil, errors.Wrapf(err, "can't open %q", inFile)
    }

    defer func() {
        f.Close()
    }()

    return Read(f, conf)
}

// Read takes a readSeeker and generates a Context,
// an in-memory representation containing a cross reference table.
func Read(rs io.ReadSeeker, conf *Configuration) (*Context, error) {

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    log.Read.Println("Read: begin")

    ctx, err := NewContext(rs, conf)
    if err != nil {
        return nil, err
    }

    if ctx.Reader15 {
        log.Info.Println("PDF Version 1.5 conforming reader")
    } else {
        log.Info.Println("PDF Version 1.4 conforming reader - no object streams or xrefstreams allowed")
    }

    // Populate xRefTable.
    if err = readXRefTable(ctx); err != nil {
        return nil, errors.Wrap(err, "Read: xRefTable failed")
    }

    // Make all objects explicitly available (load into memory) in corresponding xRefTable entries.
    // Also decode any involved object streams.
    if err = dereferenceXRefTable(ctx, conf); err != nil {
        return nil, err
    }

    // Some PDFWriters write an incorrent Size into trailer.
    if *ctx.XRefTable.Size < len(ctx.XRefTable.Table) {
        *ctx.XRefTable.Size = len(ctx.XRefTable.Table)
    }

    log.Read.Println("Read: end")

    return ctx, nil
}

// ScanLines is a split function for a Scanner that returns each line of
// text, stripped of any trailing end-of-line marker. The returned line may
// be empty. The end-of-line marker is one carriage return followed
// by one newline or one carriage return or one newline.
// The last non-empty line of input will be returned even if it has no newline.
func scanLines(data []byte, atEOF bool) (advance int, token []byte, err error) {

    if atEOF && len(data) == 0 {
        return 0, nil, nil
    }

    indCR := bytes.IndexByte(data, '\r')
    indLF := bytes.IndexByte(data, '\n')

    switch {

    case indCR >= 0 && indLF >= 0:
        if indCR < indLF {
            if indLF == indCR+1 {
                // 0x0D0A
                return indLF + 1, data[0:indCR], nil
            }
            // 0x0D ... 0x0A
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            return indCR + 1, data[0:indCR], nil
        }
        // 0x0A ... 0x0D
        return indLF + 1, data[0:indLF], nil

    case indCR >= 0:
        // We have a full carriage return terminated line.
        return indCR + 1, data[0:indCR], nil

    case indLF >= 0:
        // We have a full newline-terminated line.
        return indLF + 1, data[0:indLF], nil

    }

    // If we're at EOF, we have a final, non-terminated line. Return it.
    if atEOF {
        return len(data), data, nil
    }

    // Request more data.
    return 0, nil, nil
}

func newPositionedReader(rs io.ReadSeeker, offset *int64) (*bufio.Reader, error) {

    if _, err := rs.Seek(*offset, io.SeekStart); err != nil {
        return nil, err
    }

    log.Read.Printf("newPositionedReader: positioned to offset: %d\n", *offset)

    return bufio.NewReader(rs), nil
}

// Get the file offset of the last XRefSection.
// Go to end of file and search backwards for the first occurrence of startxref {offset} %%EOF
func offsetLastXRefSection(ctx *Context) (*int64, error) {

    rs := ctx.Read.rs

    var (
        prevBuf, workBuf []byte
        bufSize          int64 = 512
        offset           int64
    )

    for i := 1; offset == 0; i++ {

        off, err := rs.Seek(-int64(i)*bufSize, io.SeekEnd)
        if err != nil {
            return nil, errors.New("pdfcpu: can't find last xref section")
        }

        log.Read.Printf("scanning for offsetLastXRefSection starting at %d\n", off)

        curBuf := make([]byte, bufSize)

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        _, err = rs.Read(curBuf)
        if err != nil {
            return nil, err
        }

        workBuf = curBuf
        if prevBuf != nil {
            workBuf = append(curBuf, prevBuf...)
        }

        j := strings.LastIndex(string(workBuf), "startxref")
        if j == -1 {
            prevBuf = curBuf
            continue
        }

        p := workBuf[j+len("startxref"):]
        posEOF := strings.Index(string(p), "%%EOF")
        if posEOF == -1 {
            return nil, errors.New("pdfcpu: no matching %%EOF for startxref")
        }

        p = p[:posEOF]
        offset, err = strconv.ParseInt(strings.TrimSpace(string(p)), 10, 64)
        if err != nil {
            return nil, errors.New("pdfcpu: corrupted last xref section")
        }

    }

    log.Read.Printf("Offset last xrefsection: %d\n", offset)

    return &offset, nil
}

// Read next subsection entry and generate corresponding xref table entry.
func parseXRefTableEntry(s *bufio.Scanner, xRefTable *XRefTable, objectNumber int) error {

    log.Read.Println("parseXRefTableEntry: begin")

    line, err := scanLine(s)
    if err != nil {
        return err
    }

    if xRefTable.Exists(objectNumber) {
        log.Read.Printf("parseXRefTableEntry: end - Skip entry %d - already assigned\n", objectNumber)
        return nil
    }

    fields := strings.Fields(line)
    if len(fields) != 3 ||
        len(fields[0]) != 10 || len(fields[1]) != 5 || len(fields[2]) != 1 {
        return errors.New("pdfcpu: parseXRefTableEntry: corrupt xref subsection header")
    }

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    offset, err := strconv.ParseInt(fields[0], 10, 64)
    if err != nil {
        return err
    }

    generation, err := strconv.Atoi(fields[1])
    if err != nil {
        return err
    }

    entryType := fields[2]
    if entryType != "f" && entryType != "n" {
        return errors.New("pdfcpu: parseXRefTableEntry: corrupt xref subsection entry")
    }

    var xRefTableEntry XRefTableEntry

    if entryType == "n" {

        // in use object

        log.Read.Printf("parseXRefTableEntry: Object #%d is in use at offset=%d, generation=%d\n", objectNumber, offset, generation)

        if offset == 0 {
            log.Info.Printf("parseXRefTableEntry: Skip entry for in use object #%d with offset 0\n", objectNumber)
            return nil
        }

        xRefTableEntry =
            XRefTableEntry{
                Free:       false,
                Offset:     &offset,
                Generation: &generation}

    } else {

        // free object

        log.Read.Printf("parseXRefTableEntry: Object #%d is unused, next free is object#%d, generation=%d\n", objectNumber, offset, generation)

        xRefTableEntry =
            XRefTableEntry{
                Free:       true,
                Offset:     &offset,
                Generation: &generation}

    }

    log.Read.Printf("parseXRefTableEntry: Insert new xreftable entry for Object %d\n", objectNumber)

    xRefTable.Table[objectNumber] = &xRefTableEntry

    log.Read.Println("parseXRefTableEntry: end")

    return nil
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}

// Process xRef table subsection and create corrresponding xRef table entries.
func parseXRefTableSubSection(s *bufio.Scanner, xRefTable *XRefTable, fields []string) error {

    log.Read.Println("parseXRefTableSubSection: begin")

    startObjNumber, err := strconv.Atoi(fields[0])
    if err != nil {
        return err
    }

    objCount, err := strconv.Atoi(fields[1])
    if err != nil {
        return err
    }

    log.Read.Printf("detected xref subsection, startObj=%d length=%d\n", startObjNumber, objCount)

    // Process all entries of this subsection into xRefTable entries.
    for i := 0; i < objCount; i++ {
        if err = parseXRefTableEntry(s, xRefTable, startObjNumber+i); err != nil {
            return err
        }
    }

    log.Read.Println("parseXRefTableSubSection: end")

    return nil
}

// Parse compressed object.
func compressedObject(s string) (Object, error) {

    log.Read.Println("compressedObject: begin")

    o, err := parseObject(&s)
    if err != nil {
        return nil, err
    }

    d, ok := o.(Dict)
    if !ok {
        // return trivial Object: Integer, Array, etc.
        log.Read.Println("compressedObject: end, any other than dict")
        return o, nil
    }

    streamLength, streamLengthRef := d.Length()
    if streamLength == nil && streamLengthRef == nil {
        // return Dict
        log.Read.Println("compressedObject: end, dict")
        return d, nil
    }

    return nil, errors.New("pdfcpu: compressedObject: stream objects are not to be stored in an object stream")
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}

// Parse all objects of an object stream and save them into objectStreamDict.ObjArray.
func parseObjectStream(osd *ObjectStreamDict) error {

    log.Read.Printf("parseObjectStream begin: decoding %d objects.\n", osd.ObjCount)

    decodedContent := osd.Content
    prolog := decodedContent[:osd.FirstObjOffset]

    objs := strings.Fields(string(prolog))
    if len(objs)%2 > 0 {
        return errors.New("pdfcpu: parseObjectStream: corrupt object stream dict")
    }

    // e.g., 10 0 11 25 = 2 Objects: #10 @ offset 0, #11 @ offset 25

    var objArray Array

    var offsetOld int

    for i := 0; i < len(objs); i += 2 {

        offset, err := strconv.Atoi(objs[i+1])
        if err != nil {
            return err
        }

        offset += osd.FirstObjOffset

        if i > 0 {
            dstr := string(decodedContent[offsetOld:offset])
            log.Read.Printf("parseObjectStream: objString = %s\n", dstr)
            o, err := compressedObject(dstr)
            if err != nil {
                return err
            }

            log.Read.Printf("parseObjectStream: [%d] = obj %s:\n%s\n", i/2-1, objs[i-2], o)
            objArray = append(objArray, o)
        }

        if i == len(objs)-2 {
            dstr := string(decodedContent[offset:])
            log.Read.Printf("parseObjectStream: objString = %s\n", dstr)
            o, err := compressedObject(dstr)
            if err != nil {
                return err
            }

            log.Read.Printf("parseObjectStream: [%d] = obj %s:\n%s\n", i/2, objs[i], o)
            objArray = append(objArray, o)
        }

        offsetOld = offset
    }

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    osd.ObjArray = objArray

    log.Read.Println("parseObjectStream end")

    return nil
}

// For each object embedded in this xRefStream create the corresponding xRef table entry.
func extractXRefTableEntriesFromXRefStream(buf []byte, xsd *XRefStreamDict, ctx *Context) error {

    log.Read.Printf("extractXRefTableEntriesFromXRefStream begin")

    // Note:
    // A value of zero for an element in the W array indicates that the corresponding field shall not be present in the stream,
    // and the default value shall be used, if there is one.
    // If the first element is zero, the type field shall not be present, and shall default to type 1.

    i1 := xsd.W[0]
    i2 := xsd.W[1]
    i3 := xsd.W[2]

    xrefEntryLen := i1 + i2 + i3
    log.Read.Printf("extractXRefTableEntriesFromXRefStream: begin xrefEntryLen = %d\n", xrefEntryLen)

    if len(buf)%xrefEntryLen > 0 {
        return errors.New("pdfcpu: extractXRefTableEntriesFromXRefStream: corrupt xrefstream")
    }

    objCount := len(xsd.Objects)
    log.Read.Printf("extractXRefTableEntriesFromXRefStream: objCount:%d %v\n", objCount, xsd.Objects)

    log.Read.Printf("extractXRefTableEntriesFromXRefStream: len(buf):%d objCount*xrefEntryLen:%d\n", len(buf), objCount*xrefEntryLen)
    if len(buf) < objCount*xrefEntryLen {
        // Sometimes there is an additional xref entry not accounted for by "Index".
        // We ignore such a entries and do not treat this as an error.
        return errors.New("pdfcpu: extractXRefTableEntriesFromXRefStream: corrupt xrefstream")
    }

    j := 0

    // bufToInt64 interprets the content of buf as an int64.
    bufToInt64 := func(buf []byte) (i int64) {

        for _, b := range buf {
            i <<= 8
            i |= int64(b)
        }

        return
    }

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    for i := 0; i < len(buf) && j < len(xsd.Objects); i += xrefEntryLen {

        objectNumber := xsd.Objects[j]

        i2Start := i + i1
        c2 := bufToInt64(buf[i2Start : i2Start+i2])
        c3 := bufToInt64(buf[i2Start+i2 : i2Start+i2+i3])

        var xRefTableEntry XRefTableEntry

        switch buf[i] {

        case 0x00:
            // free object
            log.Read.Printf("extractXRefTableEntriesFromXRefStream: Object #%d is unused, next free is object#%d, generation=%d\n", objectNumber, c2, c3)
            g := int(c3)

            xRefTableEntry =
                XRefTableEntry{
                    Free:       true,
                    Compressed: false,
                    Offset:     &c2,
                    Generation: &g}

        case 0x01:
            // in use object
            log.Read.Printf("extractXRefTableEntriesFromXRefStream: Object #%d is in use at offset=%d, generation=%d\n", objectNumber, c2, c3)
            g := int(c3)

            xRefTableEntry =
                XRefTableEntry{
                    Free:       false,
                    Compressed: false,
                    Offset:     &c2,
                    Generation: &g}

        case 0x02:
            // compressed object
            // generation always 0.
            log.Read.Printf("extractXRefTableEntriesFromXRefStream: Object #%d is compressed at obj %5d[%d]\n", objectNumber, c2, c3)
            objNumberRef := int(c2)
            objIndex := int(c3)

            xRefTableEntry =
                XRefTableEntry{
                    Free:            false,
                    Compressed:      true,
                    ObjectStream:    &objNumberRef,
                    ObjectStreamInd: &objIndex}

            ctx.Read.ObjectStreams[objNumberRef] = true

        }

        if ctx.XRefTable.Exists(objectNumber) {
            log.Read.Printf("extractXRefTableEntriesFromXRefStream: Skip entry %d -
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already assigned\n", objectNumber)
        } else {
            ctx.Table[objectNumber] = &xRefTableEntry
        }

        j++
    }

    log.Read.Println("extractXRefTableEntriesFromXRefStream: end")

    return nil
}

func xRefStreamDict(ctx *Context, o Object, objNr int, streamOffset int64) (*XRefStreamDict, error) {

    // must be Dict
    d, ok := o.(Dict)
    if !ok {
        return nil, errors.New("pdfcpu: xRefStreamDict: no dict")
    }

    // Parse attributes for stream object.
    streamLength, streamLengthObjNr := d.Length()
    if streamLength == nil && streamLengthObjNr == nil {
        return nil, errors.New("pdfcpu: xRefStreamDict: no \"Length\" entry")
    }

    filterPipeline, err := pdfFilterPipeline(ctx, d)
    if err != nil {
        return nil, err
    }

    // We have a stream object.
    log.Read.Printf("xRefStreamDict: streamobject #%d\n", objNr)
    sd := NewStreamDict(d, streamOffset, streamLength, streamLengthObjNr, filterPipeline)

    if _, err = loadEncodedStreamContent(ctx, &sd); err != nil {
        return nil, err
    }

    // Decode xrefstream content
    if err = saveDecodedStreamContent(nil, &sd, 0, 0, true); err != nil {
        return nil, errors.Wrapf(err, "xRefStreamDict: cannot decode stream for obj#:%d\n", objNr)
    }

    return parseXRefStreamDict(&sd)
}

// Parse xRef stream and setup xrefTable entries for all embedded objects and the xref stream dict.
func parseXRefStream(rd io.Reader, offset *int64, ctx *Context) (prevOffset *int64, err error) {

    log.Read.Printf("parseXRefStream: begin at offset %d\n", *offset)

    buf, endInd, streamInd, streamOffset, err := buffer(rd)
    if err != nil {
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        return nil, err
    }

    log.Read.Printf("parseXRefStream: endInd=%[1]d(%[1]x) streamInd=%[2]d(%[2]x)\n", endInd, streamInd)

    line := string(buf)

    // We expect a stream and therefore "stream" before "endobj" if "endobj" within buffer.
    // There is no guarantee that "endobj" is contained in this buffer for large streams!
    if streamInd < 0 || (endInd > 0 && endInd < streamInd) {
        return nil, errors.New("pdfcpu: parseXRefStream: corrupt pdf file")
    }

    // Init object parse buf.
    l := line[:streamInd]

    objectNumber, generationNumber, err := parseObjectAttributes(&l)
    if err != nil {
        return nil, err
    }

    // parse this object
    log.Read.Printf("parseXRefStream: xrefstm obj#:%d gen:%d\n", *objectNumber, *generationNumber)
    log.Read.Printf("parseXRefStream: dereferencing object %d\n", *objectNumber)
    o, err := parseObject(&l)
    if err != nil {
        return nil, errors.Wrapf(err, "parseXRefStream: no object")
    }

    log.Read.Printf("parseXRefStream: we have an object: %s\n", o)

    streamOffset += *offset
    sd, err := xRefStreamDict(ctx, o, *objectNumber, streamOffset)
    if err != nil {
        return nil, err
    }
    // We have an xref stream object

    err = parseTrailerInfo(sd.Dict, ctx.XRefTable)
    if err != nil {
        return nil, err
    }

    // Parse xRefStream and create xRefTable entries for embedded objects.
    err = extractXRefTableEntriesFromXRefStream(sd.Content, sd, ctx)
    if err != nil {
        return nil, err
    }

    // Create xRefTableEntry for XRefStreamDict.
    entry :=
        XRefTableEntry{
            Free:       false,
            Offset:     offset,
            Generation: generationNumber,
            Object:     *sd}
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    log.Read.Printf("parseXRefStream: Insert new xRefTable entry for Object %d\n", *objectNumber)

    ctx.Table[*objectNumber] = &entry
    ctx.Read.XRefStreams[*objectNumber] = true
    prevOffset = sd.PreviousOffset

    log.Read.Println("parseXRefStream: end")

    return prevOffset, nil
}

// Parse an xRefStream for a hybrid PDF file.
func parseHybridXRefStream(offset *int64, ctx *Context) error {

    log.Read.Println("parseHybridXRefStream: begin")

    rd, err := newPositionedReader(ctx.Read.rs, offset)
    if err != nil {
        return err
    }

    _, err = parseXRefStream(rd, offset, ctx)
    if err != nil {
        return err
    }

    log.Read.Println("parseHybridXRefStream: end")

    return nil
}

// Parse trailer dict and return any offset of a previous xref section.
func parseTrailerInfo(d Dict, xRefTable *XRefTable) error {

    log.Read.Println("parseTrailerInfo begin")

    if _, found := d.Find("Encrypt"); found {
        encryptObjRef := d.IndirectRefEntry("Encrypt")
        if encryptObjRef != nil {
            xRefTable.Encrypt = encryptObjRef
            log.Read.Printf("parseTrailerInfo: Encrypt object: %s\n", *xRefTable.Encrypt)
        }
    }

    if xRefTable.Size == nil {
        size := d.Size()
        if size == nil {
            return errors.New("pdfcpu: parseTrailerInfo: missing entry \"Size\"")
        }
        // Not reliable!
        // Patched after all read in.
        xRefTable.Size = size
    }

    if xRefTable.Root == nil {
        rootObjRef := d.IndirectRefEntry("Root")
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        if rootObjRef == nil {
            return errors.New("pdfcpu: parseTrailerInfo: missing entry \"Root\"")
        }
        xRefTable.Root = rootObjRef
        log.Read.Printf("parseTrailerInfo: Root object: %s\n", *xRefTable.Root)
    }

    if xRefTable.Info == nil {
        infoObjRef := d.IndirectRefEntry("Info")
        if infoObjRef != nil {
            xRefTable.Info = infoObjRef
            log.Read.Printf("parseTrailerInfo: Info object: %s\n", *xRefTable.Info)
        }
    }

    if xRefTable.ID == nil {
        idArray := d.ArrayEntry("ID")
        if idArray != nil {
            xRefTable.ID = idArray
            log.Read.Printf("parseTrailerInfo: ID object: %s\n", xRefTable.ID)
        } else if xRefTable.Encrypt != nil {
            return errors.New("pdfcpu: parseTrailerInfo: missing entry \"ID\"")
        }
    }

    log.Read.Println("parseTrailerInfo end")

    return nil
}

func parseTrailerDict(trailerDict Dict, ctx *Context) (*int64, error) {

    log.Read.Println("parseTrailerDict begin")

    xRefTable := ctx.XRefTable

    err := parseTrailerInfo(trailerDict, xRefTable)
    if err != nil {
        return nil, err
    }

    if arr := trailerDict.ArrayEntry("AdditionalStreams"); arr != nil {
        log.Read.Printf("parseTrailerInfo: found AdditionalStreams: %s\n", arr)
        a := Array{}
        for _, value := range arr {
            if indRef, ok := value.(IndirectRef); ok {
                a = append(a, indRef)
            }
        }
        xRefTable.AdditionalStreams = &a
    }

    offset := trailerDict.Prev()
    if offset != nil {
        log.Read.Printf("parseTrailerDict: previous xref table section offset:%d\n", *offset)
    }

    offsetXRefStream := trailerDict.Int64Entry("XRefStm")
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    if offsetXRefStream == nil {
        // No cross reference stream.
        if !ctx.Reader15 && xRefTable.Version() >= V14 && !ctx.Read.Hybrid {
            return nil, errors.Errorf("parseTrailerDict: PDF1.4 conformant reader: found incompatible version: %s", xRefTable.VersionString())
        }
        log.Read.Println("parseTrailerDict end")
        // continue to parse previous xref section, if there is any.
        return offset, nil
    }

    // This file is using cross reference streams.

    if !ctx.Read.Hybrid {
        ctx.Read.Hybrid = true
        ctx.Read.UsingXRefStreams = true
    }

    // 1.5 conformant readers process hidden objects contained
    // in XRefStm before continuing to process any previous XRefSection.
    // Previous XRefSection is expected to have free entries for hidden entries.
    // May appear in XRefSections only.
    if ctx.Reader15 {
        if err := parseHybridXRefStream(offsetXRefStream, ctx); err != nil {
            return nil, err
        }
    }

    log.Read.Println("parseTrailerDict end")

    return offset, nil
}

func scanLineRaw(s *bufio.Scanner) (string, error) {
    if ok := s.Scan(); !ok {
        if s.Err() != nil {
            return "", s.Err()
        }
        return "", errors.New("pdfcpu: scanLineRaw: returning nothing")
    }
    return s.Text(), nil
}

func scanLine(s *bufio.Scanner) (s1 string, err error) {
    for i := 0; i <= 1; i++ {
        s1, err = scanLineRaw(s)
        if err != nil {
            return "", err
        }
        if len(s1) > 0 {
            break
        }
    }

    // Remove comment.
    i := strings.Index(s1, "%")
    if i >= 0 {
        s1 = s1[:i]
    }
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    return s1, nil
}

func isDict(s string) (bool, error) {
    o, err := parseObject(&s)
    if err != nil {
        return false, err
    }
    _, ok := o.(Dict)
    return ok, nil
}

func scanTrailer(s *bufio.Scanner, line string) (string, error) {

    var buf bytes.Buffer
    var err error
    var i, j, k int

    log.Read.Printf("line: <%s>\n", line)

    // Scan for dict start tag "<<".
    for {
        i = strings.Index(line, "<<")
        if i >= 0 {
            break
        }
        line, err = scanLine(s)
        log.Read.Printf("line: <%s>\n", line)
        if err != nil {
            return "", err
        }
    }

    line = line[i:]
    buf.WriteString(line)
    buf.WriteString(" ")
    log.Read.Printf("scanTrailer dictBuf after start tag: <%s>\n", line)

    // Scan for dict end tag ">>" but account for inner dicts.
    line = line[2:]

    for {

        if len(line) == 0 {
            line, err = scanLine(s)
            if err != nil {
                return "", err
            }
            buf.WriteString(line)
            buf.WriteString(" ")
            log.Read.Printf("scanTrailer dictBuf next line: <%s>\n", line)
        }

        i = strings.Index(line, "<<")
        if i < 0 {
            // No <<
            j = strings.Index(line, ">>")
            if j >= 0 {
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```go
                // Yes >>
                if k == 0 {
                    // Check for dict
                    ok, err := isDict(buf.String())
                    if err == nil && ok {
                        return buf.String(), nil
                    }
                } else {
                    k--
                }
                line = line[j+2:]
                continue
            }
            // No >>
            line, err = scanLine(s)
            if err != nil {
                return "", err
            }
            buf.WriteString(line)
            buf.WriteString(" ")
            log.Read.Printf("scanTrailer dictBuf next line: <%s>\n", line)
        } else {
            // Yes <<
            j = strings.Index(line, ">>")
            if j < 0 {
                // No >>
                k++
                line = line[i+2:]
            } else {
                // Yes >>
                if i < j {
                    // handle <<
                    k++
                    line = line[i+2:]
                } else {
                    // handle >>
                    if k == 0 {
                        // Check for dict
                        ok, err := isDict(buf.String())
                        if err == nil && ok {
                            return buf.String(), nil
                        }
                    } else {
                        k--
                    }
                    line = line[j+2:]
                }
            }
        }
    }
}

func processTrailer(ctx *Context, s *bufio.Scanner, line string) (*int64, error) {

    var trailerString string

    if line != "trailer" {
        trailerString = line[7:]
        log.Read.Printf("processTrailer: trailer leftover: <%s>\n", trailerString)
```

```go
    } else {
        log.Read.Printf("line (len %d) <%s>\n", len(line), line)
    }

    trailerString, err := scanTrailer(s, trailerString)
    if err != nil {
        return nil, err
    }

    log.Read.Printf("processTrailer: trailerString: (len:%d) <%s>\n", len(trailerString), trailerString)

    o, err := parseObject(&trailerString)
    if err != nil {
        return nil, err
    }

    trailerDict, ok := o.(Dict)
    if !ok {
        return nil, errors.New("pdfcpu: processTrailer: corrupt trailer dict")
    }

    log.Read.Printf("processTrailer: trailerDict:\n%s\n", trailerDict)

    return parseTrailerDict(trailerDict, ctx)
}

// Parse xRef section into corresponding number of xRef table entries.
func parseXRefSection(s *bufio.Scanner, ctx *Context) (*int64, error) {

    log.Read.Println("parseXRefSection begin")

    line, err := scanLine(s)
    if err != nil {
        return nil, err
    }

    log.Read.Printf("parseXRefSection: <%s>\n", line)

    fields := strings.Fields(line)

    // Process all sub sections of this xRef section.
    for !strings.HasPrefix(line, "trailer") && len(fields) == 2 {

        if err = parseXRefTableSubSection(s, ctx.XRefTable, fields); err != nil {
            return nil, err
        }

        // trailer or another xref table subsection ?
        if line, err = scanLine(s); err != nil {
            return nil, err
        }

        // if empty line try next line for trailer
        if len(line) == 0 {
            if line, err = scanLine(s); err != nil {
                return nil, err
            }
        }
```

```go

        fields = strings.Fields(line)
    }

    log.Read.Println("parseXRefSection: All subsections read!")

    if !strings.HasPrefix(line, "trailer") {
        return nil, errors.Errorf("xrefsection: missing trailer dict, line = <%s>", line)
    }

    log.Read.Println("parseXRefSection: parsing trailer dict..")

    return processTrailer(ctx, s, line)
}

// Get version from first line of file.
// Beginning with PDF 1.4, the Version entry in the document’s catalog dictionary
// (located via the Root entry in the file’s trailer, as described in 7.5.5, "File Trailer"),
// if present, shall be used instead of the version specified in the Header.
// Save PDF Version from header to xRefTable.
// The header version comes as the first line of the file.
// eolCount is the number of characters used for eol (1 or 2).
func headerVersion(rs io.ReadSeeker) (v *Version, eolCount int, err error) {

    log.Read.Println("headerVersion begin")

    var errCorruptHeader = errors.New("pdfcpu: headerVersion: corrupt pdf stream - no header version available")

    // Get first line of file which holds the version of this PDFFile.
    // We call this the header version.
    if _, err = rs.Seek(0, io.SeekStart); err != nil {
        return nil, 0, err
    }

    buf := make([]byte, 20)
    if _, err = rs.Read(buf); err != nil {
        return nil, 0, err
    }

    s := string(buf)
    prefix := "%PDF-"

    if len(s) < 8 || !strings.HasPrefix(s, prefix) {
        return nil, 0, errCorruptHeader
    }

    pdfVersion, err := PDFVersion(s[len(prefix) : len(prefix)+3])
    if err != nil {
        return nil, 0, errors.Wrapf(err, "headerVersion: unknown PDF Header Version")
    }

    s = s[8:]
    s = strings.TrimLeft(s, "\t\f ")

    // Detect the used eol which should be 1 (0x00, 0x0D) or 2 chars (0x0D0A)long.
    // %PDF-1.x{whiteSpace}{eol}
```

```go
    if s[0] == 0x0A {
        eolCount = 1
    } else if s[0] == 0x0D {
        eolCount = 1
        if s[9] == 0x0A {
            eolCount = 2
        }
    } else {
        return nil, 0, errCorruptHeader
    }

    log.Read.Printf("headerVersion: end, found header version: %s\n", pdfVersion)

    return &pdfVersion, eolCount, nil
}

// bypassXrefSection is a hack for digesting corrupt xref sections.
// It populates the xRefTable by reading in all indirect objects line by line
// and works on the assumption of a single xref section - meaning no incremental updates have been made.
func bypassXrefSection(ctx *Context) error {
    var z int64
    g := FreeHeadGeneration
    ctx.Table[0] = &XRefTableEntry{
        Free:       true,
        Offset:     &z,
        Generation: &g}

    rs := ctx.Read.rs
    eolCount := ctx.Read.EolCount
    var off, offset int64

    rd, err := newPositionedReader(rs, &offset)
    if err != nil {
        return err
    }

    s := bufio.NewScanner(rd)
    s.Split(scanLines)

    bb := []byte{}
    var (
        withinObj     bool
        withinXref    bool
        withinTrailer bool
    )

    for {
        line, err := scanLineRaw(s)
        if err != nil {
            break
        }
        if withinXref {
            offset += int64(len(line) + eolCount)
            if withinTrailer {
                bb = append(bb, ' ')
                bb = append(bb, line...)
                i := strings.Index(line, "startxref")
                if i >= 0 {
```

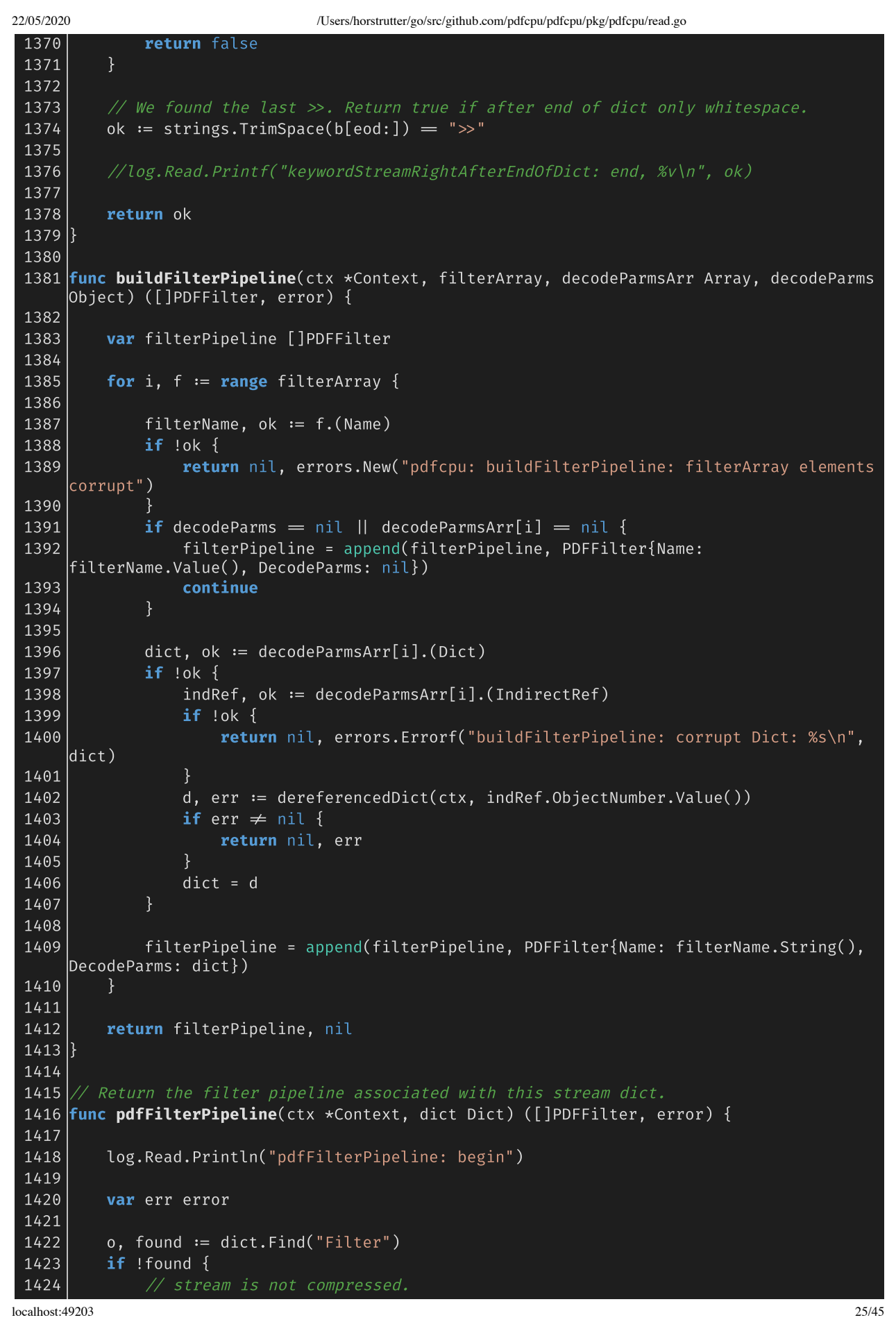
```go
                    // Parse trailer.
                    _, err = processTrailer(ctx, s, string(bb))
                    return err
                }
                continue
            }
            // Ignore all until "trailer".
            i := strings.Index(line, "trailer")
            if i >= 0 {
                bb = append(bb, line...)
                withinTrailer = true
            }
            continue
        }
        i := strings.Index(line, "xref")
        if i >= 0 {
            offset += int64(len(line) + eolCount)
            withinXref = true
            continue
        }
        if !withinObj {
            i := strings.Index(line, "obj")
            if i >= 0 {
                withinObj = true
                off = offset
                bb = append(bb, line[:i+3]...)
            }
            offset += int64(len(line) + eolCount)
            continue
        }

        // within obj
        offset += int64(len(line) + eolCount)
        bb = append(bb, ' ')
        bb = append(bb, line...)
        i = strings.Index(line, "endobj")
        if i >= 0 {
            l := string(bb)
            objNr, generation, err := parseObjectAttributes(&l)
            if err != nil {
                return err
            }
            of := off
            ctx.Table[*objNr] = &XRefTableEntry{
                Free:       false,
                Offset:     &of,
                Generation: generation}
            bb = nil
            withinObj = false
        }
    }
    return nil
}

// Build XRefTable by reading XRef streams or XRef sections.
func buildXRefTableStartingAt(ctx *Context, offset *int64) error {

    log.Read.Println("buildXRefTableStartingAt: begin")

```

```go
    rs := ctx.Read.rs

    hv, eolCount, err := headerVersion(rs)
    if err != nil {
        return err
    }

    ctx.HeaderVersion = hv
    ctx.Read.EolCount = eolCount

    for offset != nil {

        rd, err := newPositionedReader(rs, offset)
        if err != nil {
            return err
        }

        s := bufio.NewScanner(rd)
        s.Split(scanLines)

        line, err := scanLine(s)
        if err != nil {
            return err
        }

        log.Read.Printf("line: <%s>\n", line)

        if strings.TrimSpace(line) == "xref" {
            log.Read.Println("buildXRefTableStartingAt: found xref section")
            if offset, err = parseXRefSection(s, ctx); err != nil {
                return err
            }
        } else {

            log.Read.Println("buildXRefTableStartingAt: found xref stream")
            ctx.Read.UsingXRefStreams = true
            rd, err = newPositionedReader(rs, offset)
            if err != nil {
                return err
            }
            if offset, err = parseXRefStream(rd, offset, ctx); err != nil {
                log.Read.Printf("bypassXRefSection after %v\n", err)
                // Try fix for corrupt single xref section.
                return bypassXrefSection(ctx)
            }
        }
    }

    log.Read.Println("buildXRefTableStartingAt: end")

    return nil
}

// Populate the cross reference table for this PDF file.
// Goto offset of first xref table entry.
// Can be "xref" or indirect object reference eg. "34 0 obj"
// Keep digesting xref sections as long as there is a defined previous xref section
// and build up the xref table along the way.
func readXRefTable(ctx *Context) (err error) {
```

```go

    log.Read.Println("readXRefTable: begin")

    offset, err := offsetLastXRefSection(ctx)
    if err != nil {
        return
    }

    err = buildXRefTableStartingAt(ctx, offset)
    if err == io.EOF {
        return errors.Wrap(err, "readXRefTable: unexpected eof")
    }
    if err != nil {
        return
    }

    // Log list of free objects (not the "free list").
    //log.Read.Printf("freelist: %v\n", ctx.FreeObjects)

    // Ensure valid freelist of objects.
    err = ctx.EnsureValidFreeList()
    if err != nil {
        return
    }

    log.Read.Println("readXRefTable: end")

    return
}

func growBufBy(buf []byte, size int, rd io.Reader) ([]byte, error) {

    b := make([]byte, size)

    _, err := rd.Read(b)
    if err != nil {
        return nil, err
    }
    //log.Read.Printf("growBufBy: Read %d bytes\n", n)

    return append(buf, b...), nil
}

func nextStreamOffset(line string, streamInd int) (off int) {

    off = streamInd + len("stream")

    // Skip optional blanks.
    // TODO Should be skip optional whitespace instead?
    for ; line[off] == 0x20; off++ {
    }

    // Skip 0A eol.
    if line[off] == '\n' {
        off++
        return
    }

    // Skip 0D eol.
```

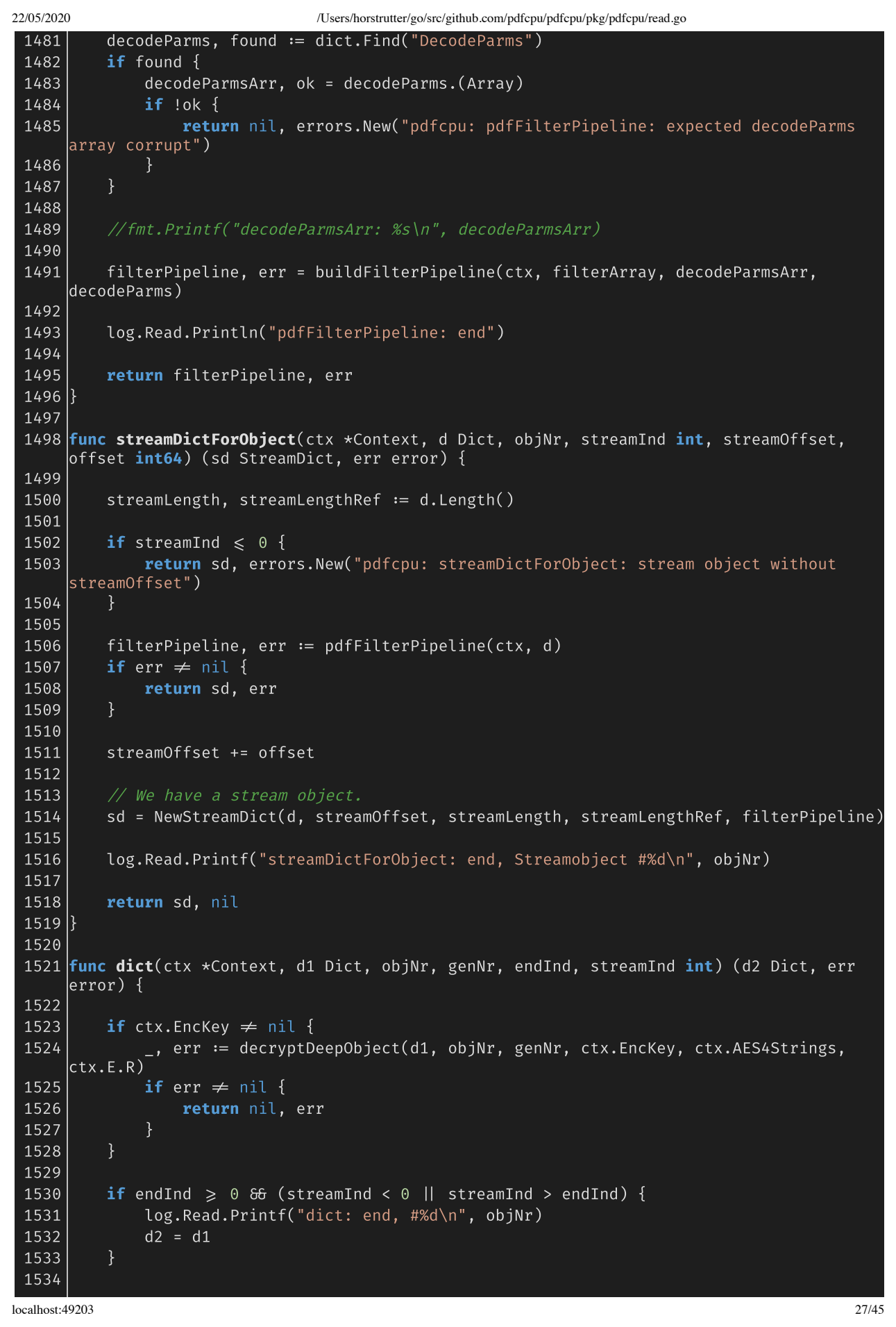
```go
    if line[off] == '\r' {
        off++
        // Skip 0D0A eol.
        if line[off] == '\n' {
            off++
        }
    }

    return
}

func lastStreamMarker(streamInd *int, endInd int, line string) {

    if *streamInd > len(line)-len("stream") {
        // No space for another stream marker.
        *streamInd = -1
        return
    }

    // We start searching after this stream marker.
    bufpos := *streamInd + len("stream")

    // Search for next stream marker.
    i := strings.Index(line[bufpos:], "stream")
    if i < 0 {
        // No stream marker within line buffer.
        *streamInd = -1
        return
    }

    // We found the next stream marker.
    *streamInd += len("stream") + i

    if endInd > 0 && *streamInd > endInd {
        // We found a stream marker of another object
        *streamInd = -1
    }

}

// Provide a PDF file buffer of sufficient size for parsing an object w/o stream.
func buffer(rd io.Reader) (buf []byte, endInd int, streamInd int, streamOffset int64, err error) {

    // process: # gen obj ... obj dict ... {stream ... data ... endstream} ... endobj
    //                                      streamInd                          endInd
    //                                    -1 if absent                       -1 if absent

    //log.Read.Println("buffer: begin")

    endInd, streamInd = -1, -1

    for endInd < 0 && streamInd < 0 {

        buf, err = growBufBy(buf, defaultBufSize, rd)
        if err != nil {
            return nil, 0, 0, 0, err
        }
```

```go

        line := string(buf)
        endInd = strings.Index(line, "endobj")
        streamInd = strings.Index(line, "stream")

        if endInd > 0 && (streamInd < 0 || streamInd > endInd) {
            // No stream marker in buf detected.
            break
        }

        // For very rare cases where "stream" also occurs within obj dict
        // we need to find the last "stream" marker before a possible end marker.
        for streamInd > 0 && !keywordStreamRightAfterEndOfDict(line, streamInd) {
            lastStreamMarker(&streamInd, endInd, line)
        }

        log.Read.Printf("buffer: endInd=%d streamInd=%d\n", endInd, streamInd)

        if streamInd > 0 {

            // streamOffset ... the offset where the actual stream data begins.
            //                  is right after the eol after "stream".

            slack := 10 // for optional whitespace + eol (max 2 chars)
            need := streamInd + len("stream") + slack

            if len(line) < need {

                // to prevent buffer overflow.
                buf, err = growBufBy(buf, need-len(line), rd)
                if err != nil {
                    return nil, 0, 0, 0, err
                }

                line = string(buf)
            }

            streamOffset = int64(nextStreamOffset(line, streamInd))
        }
    }

    //log.Read.Printf("buffer: end, returned bufsize=%d streamOffset=%d\n", len(buf), streamOffset)

    return buf, endInd, streamInd, streamOffset, nil
}

// return true if 'stream' follows end of dict: >>{whitespace}stream
func keywordStreamRightAfterEndOfDict(buf string, streamInd int) bool {

    //log.Read.Println("keywordStreamRightAfterEndOfDict: begin")

    // Get a slice of the chunk right in front of 'stream'.
    b := buf[:streamInd]

    // Look for last end of dict marker.
    eod := strings.LastIndex(b, ">>")
    if eod < 0 {
        // No end of dict in buf.
```

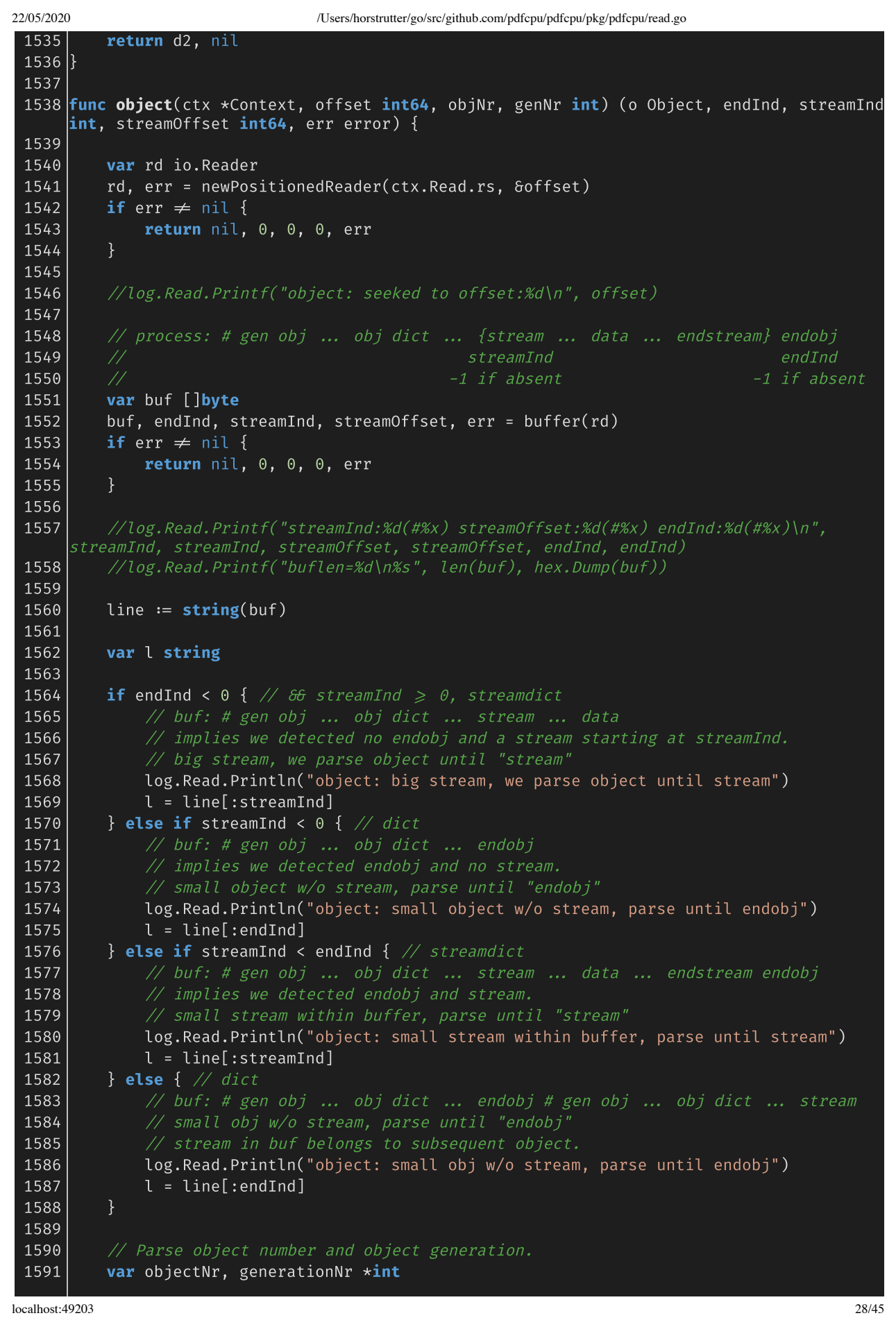


```go
        return false
    }

    // We found the last >>. Return true if after end of dict only whitespace.
    ok := strings.TrimSpace(b[eod:]) == ">>"

    //log.Read.Printf("keywordStreamRightAfterEndOfDict: end, %v\n", ok)

    return ok
}

func buildFilterPipeline(ctx *Context, filterArray, decodeParmsArr Array, decodeParms Object) ([]PDFFilter, error) {

    var filterPipeline []PDFFilter

    for i, f := range filterArray {

        filterName, ok := f.(Name)
        if !ok {
            return nil, errors.New("pdfcpu: buildFilterPipeline: filterArray elements corrupt")
        }
        if decodeParms == nil || decodeParmsArr[i] == nil {
            filterPipeline = append(filterPipeline, PDFFilter{Name: filterName.Value(), DecodeParms: nil})
            continue
        }

        dict, ok := decodeParmsArr[i].(Dict)
        if !ok {
            indRef, ok := decodeParmsArr[i].(IndirectRef)
            if !ok {
                return nil, errors.Errorf("buildFilterPipeline: corrupt Dict: %s\n", dict)
            }
            d, err := dereferencedDict(ctx, indRef.ObjectNumber.Value())
            if err != nil {
                return nil, err
            }
            dict = d
        }

        filterPipeline = append(filterPipeline, PDFFilter{Name: filterName.String(), DecodeParms: dict})
    }

    return filterPipeline, nil
}

// Return the filter pipeline associated with this stream dict.
func pdfFilterPipeline(ctx *Context, dict Dict) ([]PDFFilter, error) {

    log.Read.Println("pdfFilterPipeline: begin")

    var err error

    o, found := dict.Find("Filter")
    if !found {
        // stream is not compressed.
```

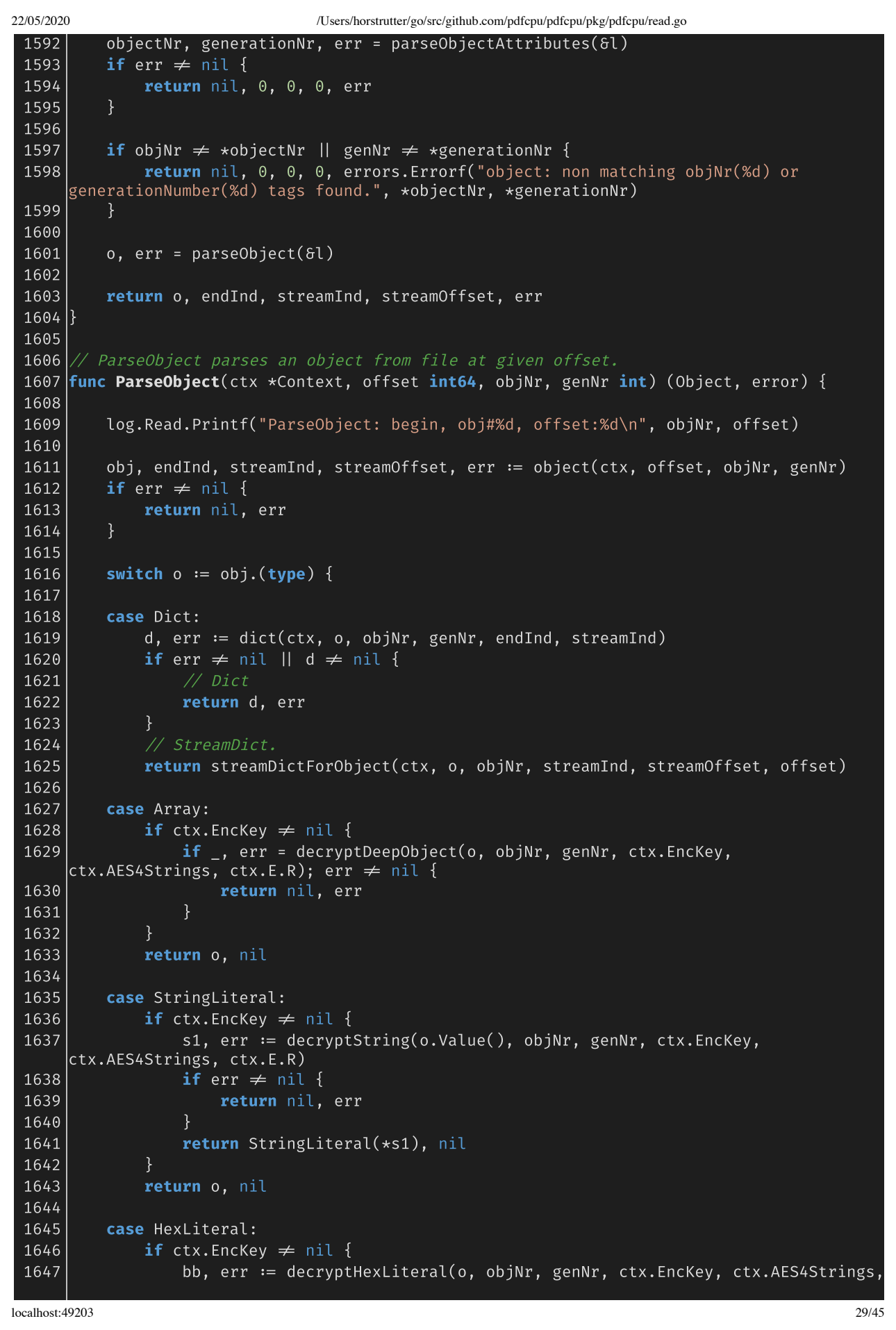
```go
        return nil, nil
    }

    // compressed stream.

    var filterPipeline []PDFFilter

    if indRef, ok := o.(IndirectRef); ok {
        o, err = dereferencedObject(ctx, indRef.ObjectNumber.Value())
        if err != nil {
            return nil, err
        }
    }

    //fmt.Printf("dereferenced filter obj: %s\n", obj)

    if name, ok := o.(Name); ok {

        // single filter.

        filterName := name.String()

        o, found := dict.Find("DecodeParms")
        if !found {
            // w/o decode parameters.
            log.Read.Println("pdfFilterPipeline: end w/o decode parms")
            return append(filterPipeline, PDFFilter{Name: filterName, DecodeParms: nil}), nil
        }

        d, ok := o.(Dict)
        if !ok {
            ir, ok := o.(IndirectRef)
            if !ok {
                return nil, errors.Errorf("pdfFilterPipeline: corrupt Dict: %s\n", o)
            }
            d, err = dereferencedDict(ctx, ir.ObjectNumber.Value())
            if err != nil {
                return nil, err
            }
        }

        // with decode parameters.
        log.Read.Println("pdfFilterPipeline: end with decode parms")
        return append(filterPipeline, PDFFilter{Name: filterName, DecodeParms: d}), nil
    }

    // filter pipeline.

    // Array of filternames
    filterArray, ok := o.(Array)
    if !ok {
        return nil, errors.Errorf("pdfFilterPipeline: Expected filterArray corrupt, %v %T", o, o)
    }

    // Optional array of decode parameter dicts.
    var decodeParmsArr Array
```



```go
    decodeParms, found := dict.Find("DecodeParms")
    if found {
        decodeParmsArr, ok = decodeParms.(Array)
        if !ok {
            return nil, errors.New("pdfcpu: pdfFilterPipeline: expected decodeParms array corrupt")
        }
    }

    //fmt.Printf("decodeParmsArr: %s\n", decodeParmsArr)

    filterPipeline, err = buildFilterPipeline(ctx, filterArray, decodeParmsArr, decodeParms)

    log.Read.Println("pdfFilterPipeline: end")

    return filterPipeline, err
}

func streamDictForObject(ctx *Context, d Dict, objNr, streamInd int, streamOffset, offset int64) (sd StreamDict, err error) {

    streamLength, streamLengthRef := d.Length()

    if streamInd <= 0 {
        return sd, errors.New("pdfcpu: streamDictForObject: stream object without streamOffset")
    }

    filterPipeline, err := pdfFilterPipeline(ctx, d)
    if err != nil {
        return sd, err
    }

    streamOffset += offset

    // We have a stream object.
    sd = NewStreamDict(d, streamOffset, streamLength, streamLengthRef, filterPipeline)

    log.Read.Printf("streamDictForObject: end, Streamobject #%d\n", objNr)

    return sd, nil
}

func dict(ctx *Context, d1 Dict, objNr, genNr, endInd, streamInd int) (d2 Dict, err error) {

    if ctx.EncKey != nil {
        _, err := decryptDeepObject(d1, objNr, genNr, ctx.EncKey, ctx.AES4Strings, ctx.E.R)
        if err != nil {
            return nil, err
        }
    }

    if endInd >= 0 && (streamInd < 0 || streamInd > endInd) {
        log.Read.Printf("dict: end, #%d\n", objNr)
        d2 = d1
    }

```

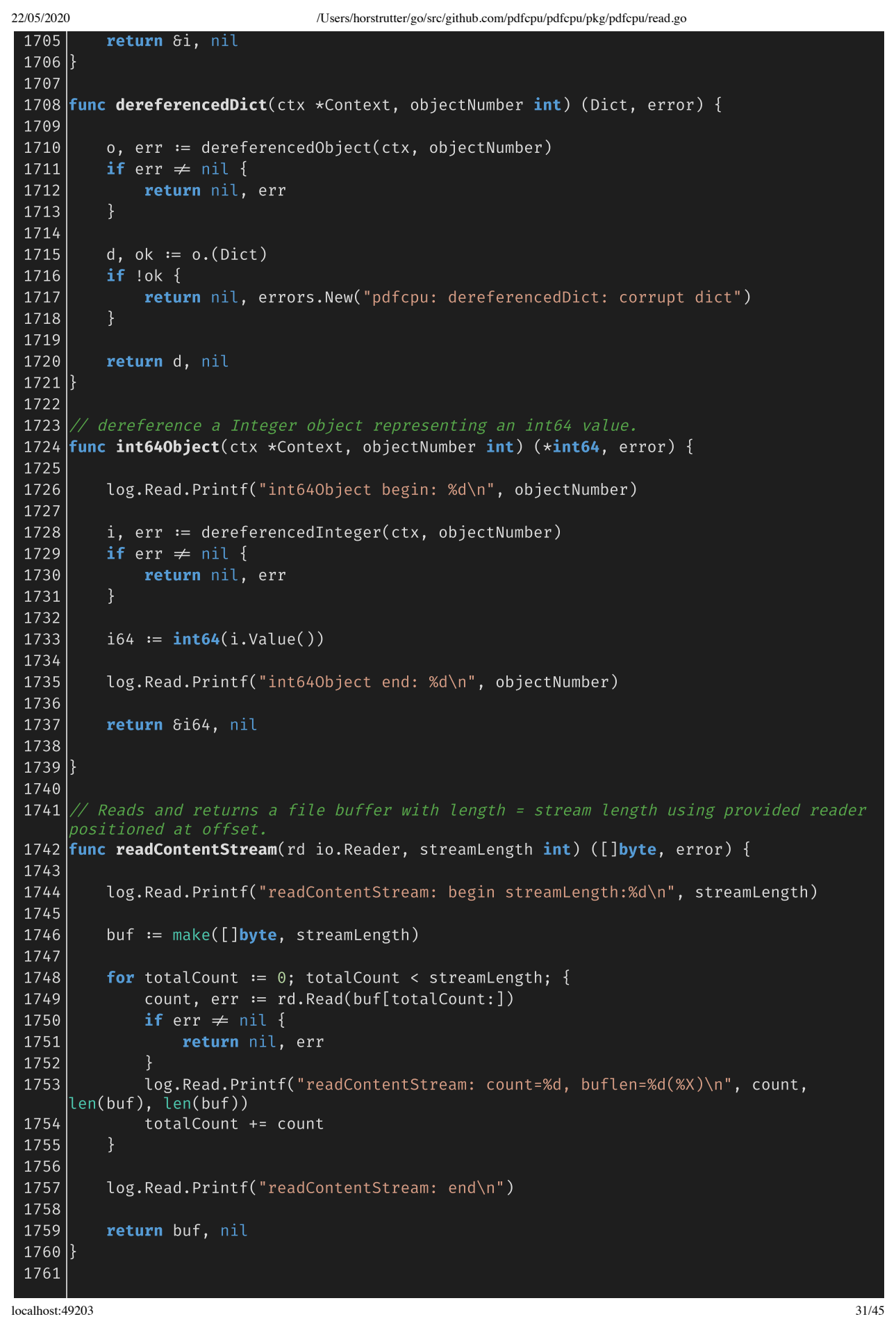


```go
    return d2, nil
}

func object(ctx *Context, offset int64, objNr, genNr int) (o Object, endInd, streamInd int, streamOffset int64, err error) {

    var rd io.Reader
    rd, err = newPositionedReader(ctx.Read.rs, &offset)
    if err != nil {
        return nil, 0, 0, 0, err
    }

    //log.Read.Printf("object: seeked to offset:%d\n", offset)

    // process: # gen obj ... obj dict ... {stream ... data ... endstream} endobj
    //                                    streamInd                        endInd
    //                                  -1 if absent                    -1 if absent
    var buf []byte
    buf, endInd, streamInd, streamOffset, err = buffer(rd)
    if err != nil {
        return nil, 0, 0, 0, err
    }

    //log.Read.Printf("streamInd:%d(#%x) streamOffset:%d(#%x) endInd:%d(#%x)\n", streamInd, streamInd, streamOffset, streamOffset, endInd, endInd)
    //log.Read.Printf("buflen=%d\n%s", len(buf), hex.Dump(buf))

    line := string(buf)

    var l string

    if endInd < 0 { // && streamInd >= 0, streamdict
        // buf: # gen obj ... obj dict ... stream ... data
        // implies we detected no endobj and a stream starting at streamInd.
        // big stream, we parse object until "stream"
        log.Read.Println("object: big stream, we parse object until stream")
        l = line[:streamInd]
    } else if streamInd < 0 { // dict
        // buf: # gen obj ... obj dict ... endobj
        // implies we detected endobj and no stream.
        // small object w/o stream, parse until "endobj"
        log.Read.Println("object: small object w/o stream, parse until endobj")
        l = line[:endInd]
    } else if streamInd < endInd { // streamdict
        // buf: # gen obj ... obj dict ... stream ... data ... endstream endobj
        // implies we detected endobj and stream.
        // small stream within buffer, parse until "stream"
        log.Read.Println("object: small stream within buffer, parse until stream")
        l = line[:streamInd]
    } else { // dict
        // buf: # gen obj ... obj dict ... endobj # gen obj ... obj dict ... stream
        // small obj w/o stream, parse until "endobj"
        // stream in buf belongs to subsequent object.
        log.Read.Println("object: small obj w/o stream, parse until endobj")
        l = line[:endInd]
    }

    // Parse object number and object generation.
    var objectNr, generationNr *int
```

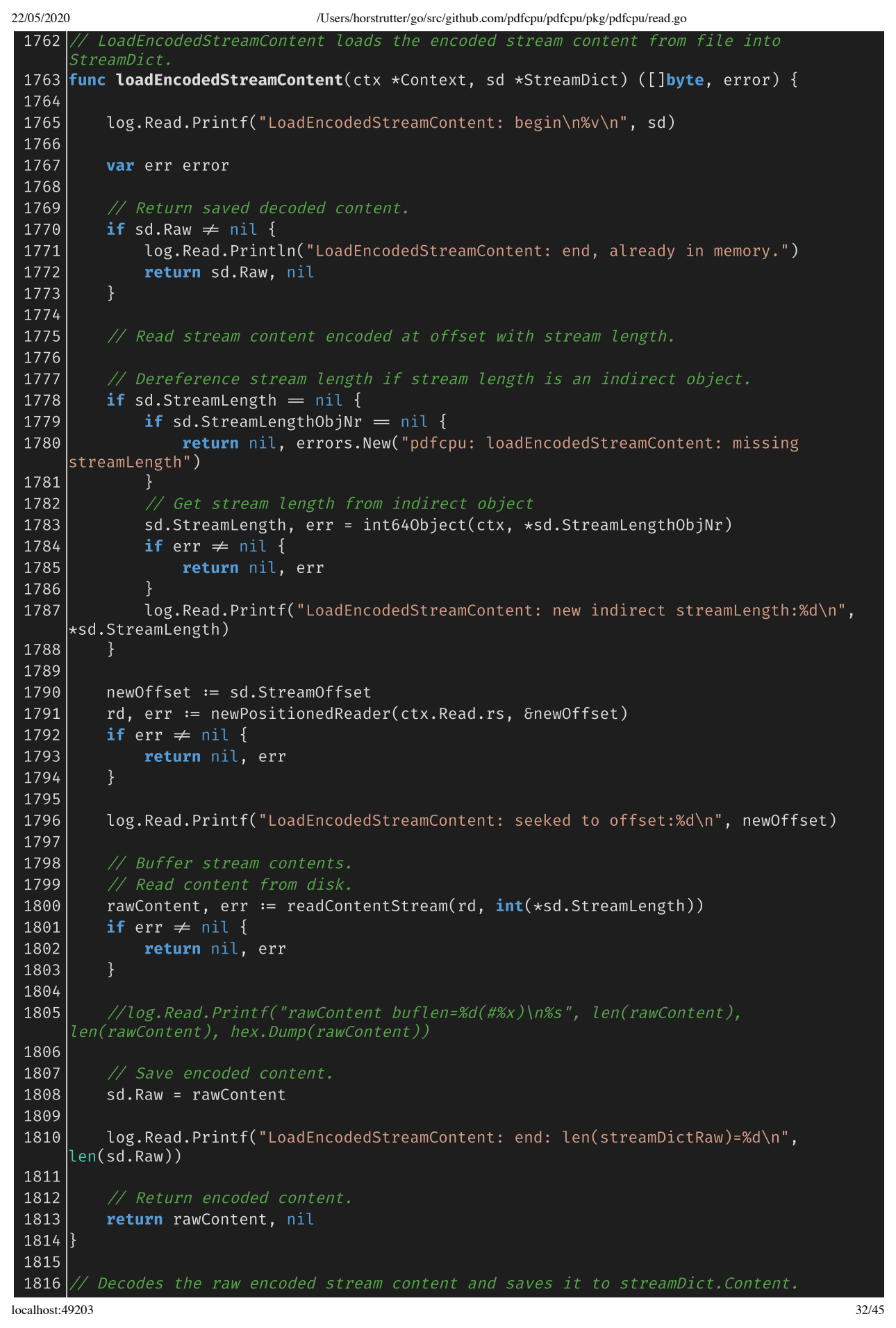


```go
    objectNr, generationNr, err = parseObjectAttributes(&l)
    if err != nil {
        return nil, 0, 0, 0, err
    }

    if objNr != *objectNr || genNr != *generationNr {
        return nil, 0, 0, 0, errors.Errorf("object: non matching objNr(%d) or generationNumber(%d) tags found.", *objectNr, *generationNr)
    }

    o, err = parseObject(&l)

    return o, endInd, streamInd, streamOffset, err
}

// ParseObject parses an object from file at given offset.
func ParseObject(ctx *Context, offset int64, objNr, genNr int) (Object, error) {

    log.Read.Printf("ParseObject: begin, obj#%d, offset:%d\n", objNr, offset)

    obj, endInd, streamInd, streamOffset, err := object(ctx, offset, objNr, genNr)
    if err != nil {
        return nil, err
    }

    switch o := obj.(type) {

    case Dict:
        d, err := dict(ctx, o, objNr, genNr, endInd, streamInd)
        if err != nil || d != nil {
            // Dict
            return d, err
        }
        // StreamDict.
        return streamDictForObject(ctx, o, objNr, streamInd, streamOffset, offset)

    case Array:
        if ctx.EncKey != nil {
            if _, err = decryptDeepObject(o, objNr, genNr, ctx.EncKey, ctx.AES4Strings, ctx.E.R); err != nil {
                return nil, err
            }
        }
        return o, nil

    case StringLiteral:
        if ctx.EncKey != nil {
            s1, err := decryptString(o.Value(), objNr, genNr, ctx.EncKey, ctx.AES4Strings, ctx.E.R)
            if err != nil {
                return nil, err
            }
            return StringLiteral(*s1), nil
        }
        return o, nil

    case HexLiteral:
        if ctx.EncKey != nil {
            bb, err := decryptHexLiteral(o, objNr, genNr, ctx.EncKey, ctx.AES4Strings,
```

```go
ctx.E.R)
            if err != nil {
                return nil, err
            }
            return StringLiteral(string(bb)), nil
        }
        return o, nil

    default:
        return o, nil
    }
}

func dereferencedObject(ctx *Context, objectNumber int) (Object, error) {

    entry, ok := ctx.Find(objectNumber)
    if !ok {
        return nil, errors.New("pdfcpu: dereferencedObject: unregistered object")
    }

    if entry.Compressed {
        err := decompressXRefTableEntry(ctx.XRefTable, objectNumber, entry)
        if err != nil {
            return nil, err
        }
    }

    if entry.Object == nil {

        log.Read.Printf("dereferencedObject: dereferencing object %d\n", objectNumber)

        o, err := ParseObject(ctx, *entry.Offset, objectNumber, *entry.Generation)
        if err != nil {
            return nil, errors.Wrapf(err, "dereferencedObject: problem dereferencing object %d", objectNumber)
        }

        if o == nil {
            return nil, errors.New("pdfcpu: dereferencedObject: object is nil")
        }

        entry.Object = o
    }

    return entry.Object, nil
}

func dereferencedInteger(ctx *Context, objectNumber int) (*Integer, error) {

    o, err := dereferencedObject(ctx, objectNumber)
    if err != nil {
        return nil, err
    }

    i, ok := o.(Integer)
    if !ok {
        return nil, errors.New("pdfcpu: dereferencedInteger: corrupt integer")
    }

```



```go
    return &i, nil
}

func dereferencedDict(ctx *Context, objectNumber int) (Dict, error) {

    o, err := dereferencedObject(ctx, objectNumber)
    if err != nil {
        return nil, err
    }

    d, ok := o.(Dict)
    if !ok {
        return nil, errors.New("pdfcpu: dereferencedDict: corrupt dict")
    }

    return d, nil
}

// dereference a Integer object representing an int64 value.
func int64Object(ctx *Context, objectNumber int) (*int64, error) {

    log.Read.Printf("int64Object begin: %d\n", objectNumber)

    i, err := dereferencedInteger(ctx, objectNumber)
    if err != nil {
        return nil, err
    }

    i64 := int64(i.Value())

    log.Read.Printf("int64Object end: %d\n", objectNumber)

    return &i64, nil

}

// Reads and returns a file buffer with length = stream length using provided reader positioned at offset.
func readContentStream(rd io.Reader, streamLength int) ([]byte, error) {

    log.Read.Printf("readContentStream: begin streamLength:%d\n", streamLength)

    buf := make([]byte, streamLength)

    for totalCount := 0; totalCount < streamLength; {
        count, err := rd.Read(buf[totalCount:])
        if err != nil {
            return nil, err
        }
        log.Read.Printf("readContentStream: count=%d, buflen=%d(%X)\n", count, len(buf), len(buf))
        totalCount += count
    }

    log.Read.Printf("readContentStream: end\n")

    return buf, nil
}

```

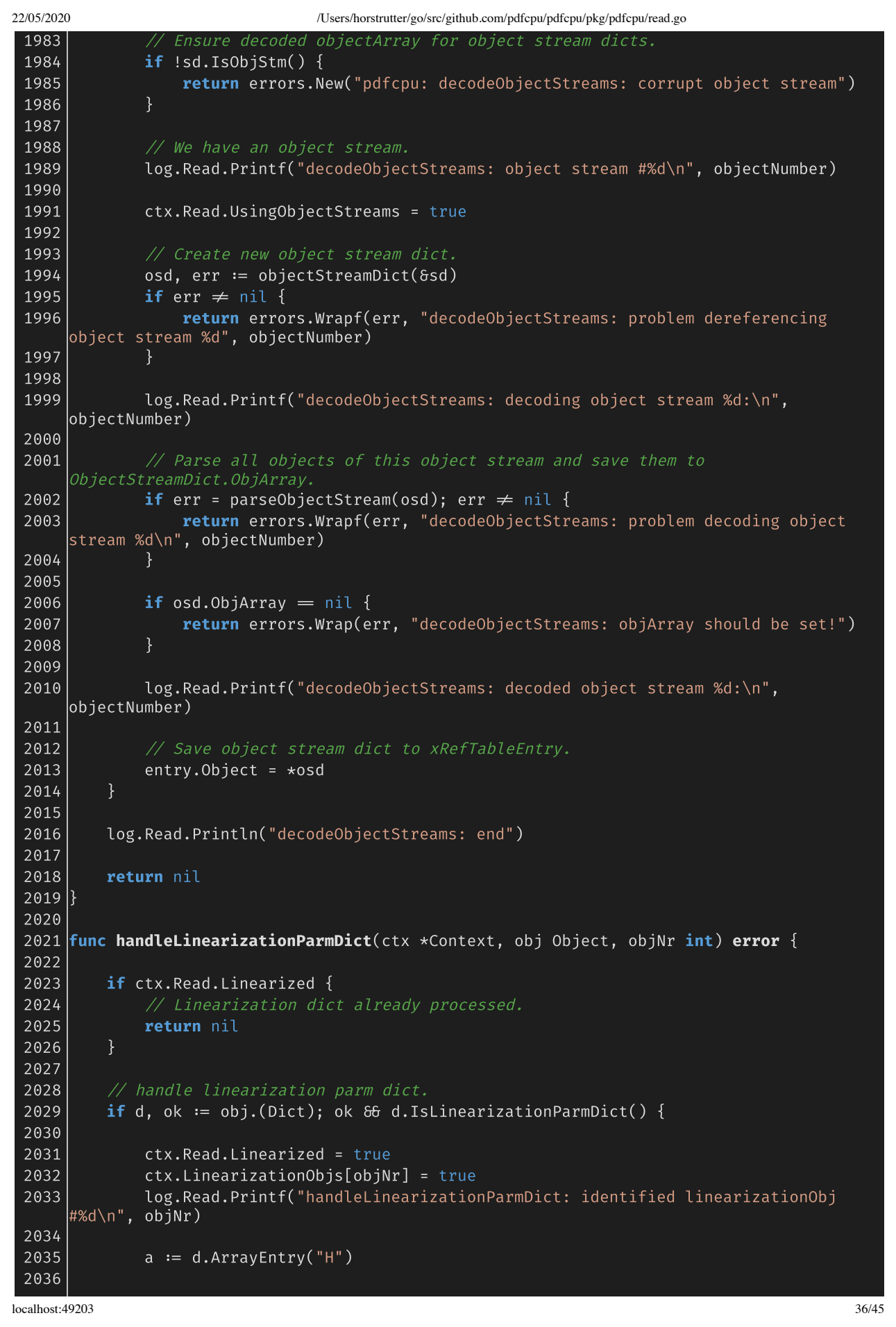


```go
// LoadEncodedStreamContent loads the encoded stream content from file into StreamDict.
func loadEncodedStreamContent(ctx *Context, sd *StreamDict) ([]byte, error) {

    log.Read.Printf("LoadEncodedStreamContent: begin\n%v\n", sd)

    var err error

    // Return saved decoded content.
    if sd.Raw != nil {
        log.Read.Println("LoadEncodedStreamContent: end, already in memory.")
        return sd.Raw, nil
    }

    // Read stream content encoded at offset with stream length.

    // Dereference stream length if stream length is an indirect object.
    if sd.StreamLength == nil {
        if sd.StreamLengthObjNr == nil {
            return nil, errors.New("pdfcpu: loadEncodedStreamContent: missing streamLength")
        }
        // Get stream length from indirect object
        sd.StreamLength, err = int64Object(ctx, *sd.StreamLengthObjNr)
        if err != nil {
            return nil, err
        }
        log.Read.Printf("LoadEncodedStreamContent: new indirect streamLength:%d\n", *sd.StreamLength)
    }

    newOffset := sd.StreamOffset
    rd, err := newPositionedReader(ctx.Read.rs, &newOffset)
    if err != nil {
        return nil, err
    }

    log.Read.Printf("LoadEncodedStreamContent: seeked to offset:%d\n", newOffset)

    // Buffer stream contents.
    // Read content from disk.
    rawContent, err := readContentStream(rd, int(*sd.StreamLength))
    if err != nil {
        return nil, err
    }

    //log.Read.Printf("rawContent buflen=%d(#%x)\n%s", len(rawContent), len(rawContent), hex.Dump(rawContent))

    // Save encoded content.
    sd.Raw = rawContent

    log.Read.Printf("LoadEncodedStreamContent: end: len(streamDictRaw)=%d\n", len(sd.Raw))

    // Return encoded content.
    return rawContent, nil
}

// Decodes the raw encoded stream content and saves it to streamDict.Content.
```

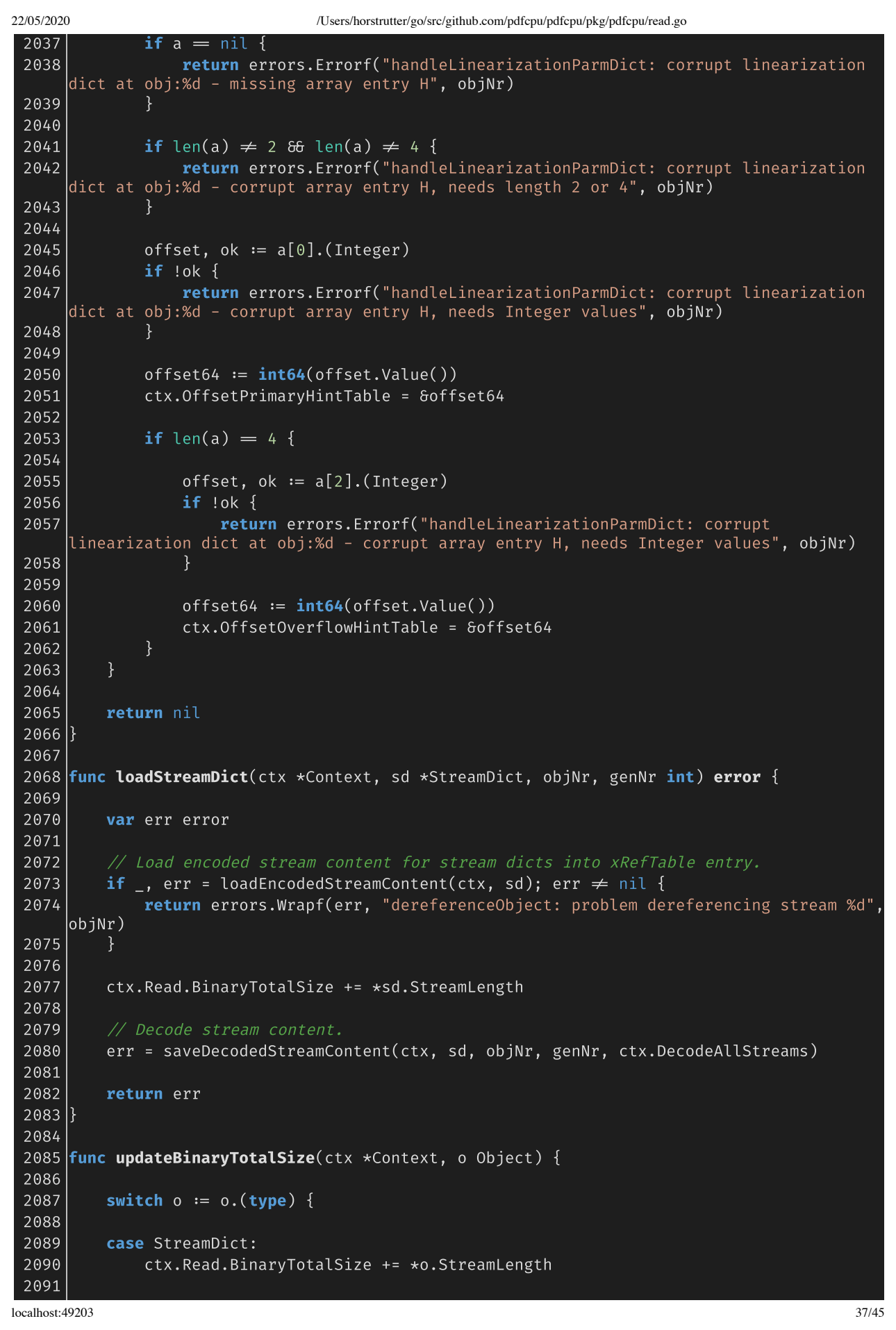
```go
func saveDecodedStreamContent(ctx *Context, sd *StreamDict, objNr, genNr int, decode bool) (err error) {

    log.Read.Printf("saveDecodedStreamContent: begin decode=%t\n", decode)

    // If the "Identity" crypt filter is used we do not need to decrypt.
    if ctx != nil && ctx.EncKey != nil {
        if len(sd.FilterPipeline) == 1 && sd.FilterPipeline[0].Name == "Crypt" {
            sd.Content = sd.Raw
            return nil
        }
    }

    // Special case: If the length of the encoded data is 0, we do not need to decode anything.
    if len(sd.Raw) == 0 {
        sd.Content = sd.Raw
        return nil
    }

    // ctx gets created after XRefStream parsing.
    // XRefStreams are not encrypted.
    if ctx != nil && ctx.EncKey != nil {
        sd.Raw, err = decryptStream(sd.Raw, objNr, genNr, ctx.EncKey, ctx.AES4Streams, ctx.E.R)
        if err != nil {
            return err
        }
        l := int64(len(sd.Raw))
        sd.StreamLength = &l
    }

    if !decode {
        return nil
    }

    // Actual decoding of content stream.
    err = decodeStream(sd)
    if err == filter.ErrUnsupportedFilter {
        err = nil
    }
    if err != nil {
        return err
    }

    log.Read.Println("saveDecodedStreamContent: end")

    return nil
}

// Resolve compressed xRefTableEntry
func decompressXRefTableEntry(xRefTable *XRefTable, objectNumber int, entry *XRefTableEntry) error {

    log.Read.Printf("decompressXRefTableEntry: compressed object %d at %d[%d]\n", objectNumber, *entry.ObjectStream, *entry.ObjectStreamInd)

    // Resolve xRefTable entry of referenced object stream.
    objectStreamXRefTableEntry, ok := xRefTable.Find(*entry.ObjectStream)
    if !ok {
```

```go
        return errors.Errorf("decompressXRefTableEntry: problem dereferencing object stream %d, no xref table entry", *entry.ObjectStream)
    }

    // Object of this entry has to be a ObjectStreamDict.
    sd, ok := objectStreamXRefTableEntry.Object.(ObjectStreamDict)
    if !ok {
        return errors.Errorf("decompressXRefTableEntry: problem dereferencing object stream %d, no object stream", *entry.ObjectStream)
    }

    // Get indexed object from ObjectStreamDict.
    o, err := sd.IndexedObject(*entry.ObjectStreamInd)
    if err != nil {
        return errors.Wrapf(err, "decompressXRefTableEntry: problem dereferencing object stream %d", *entry.ObjectStream)
    }

    // Save object to XRefRableEntry.
    g := 0
    entry.Object = o
    entry.Generation = &g
    entry.Compressed = false

    log.Read.Printf("decompressXRefTableEntry: end, Obj %d[%d]:\n<%s>\n", *entry.ObjectStream, *entry.ObjectStreamInd, o)

    return nil
}

// Log interesting stream content.
func logStream(o Object) {

    switch o := o.(type) {

    case StreamDict:

        if o.Content == nil {
            log.Read.Println("logStream: no stream content")
        }

        if o.IsPageContent {
            //log.Read.Printf("content <%s>\n", StreamDict.Content)
        }

    case ObjectStreamDict:

        if o.Content == nil {
            log.Read.Println("logStream: no object stream content")
        } else {
            log.Read.Printf("logStream: objectStream content = %s\n", o.Content)
        }

        if o.ObjArray == nil {
            log.Read.Println("logStream: no object stream obj arr")
        } else {
            log.Read.Printf("logStream: objectStream objArr = %s\n", o.ObjArray)
        }

    default:
```

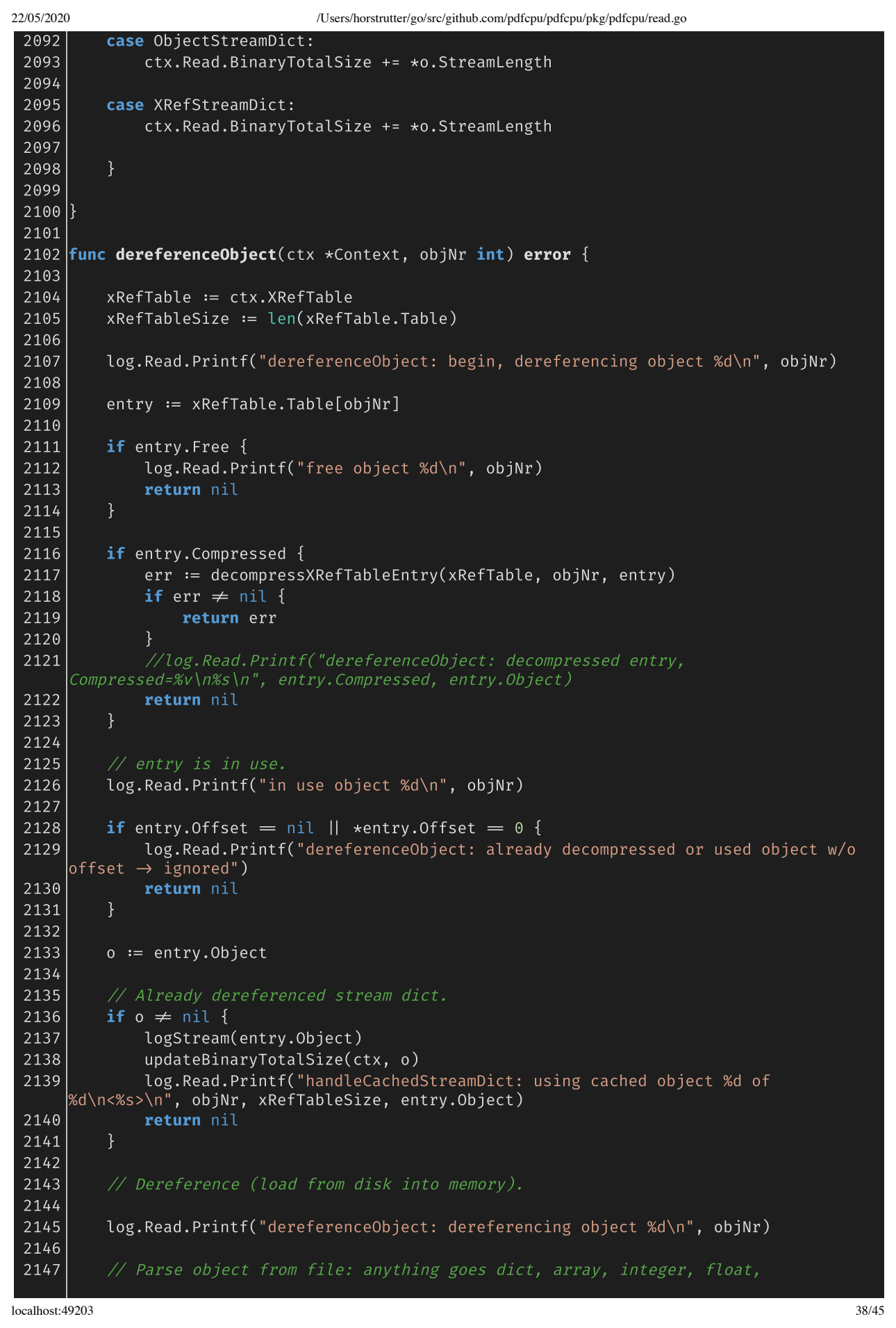
```go
        log.Read.Println("logStream: no ObjectStreamDict")

    }

}

// Decode all object streams so contained objects are ready to be used.
func decodeObjectStreams(ctx *Context) error {

    // Note:
    // Entry "Extends" intentionally left out.
    // No object stream collection validation necessary.

    log.Read.Println("decodeObjectStreams: begin")

    // Get sorted slice of object numbers.
    var keys []int
    for k := range ctx.Read.ObjectStreams {
        keys = append(keys, k)
    }
    sort.Ints(keys)

    for _, objectNumber := range keys {

        // Get XRefTableEntry.
        entry := ctx.XRefTable.Table[objectNumber]
        if entry == nil {
            return errors.Errorf("decodeObjectStream: missing entry for obj#%d\n", objectNumber)
        }

        log.Read.Printf("decodeObjectStreams: parsing object stream for obj#%d\n", objectNumber)

        // Parse object stream from file.
        o, err := ParseObject(ctx, *entry.Offset, objectNumber, *entry.Generation)
        if err != nil || o == nil {
            return errors.New("pdfcpu: decodeObjectStreams: corrupt object stream")
        }

        // Ensure StreamDict
        sd, ok := o.(StreamDict)
        if !ok {
            return errors.New("pdfcpu: decodeObjectStreams: corrupt object stream")
        }

        // Load encoded stream content to xRefTable.
        if _, err = loadEncodedStreamContent(ctx, &sd); err != nil {
            return errors.Wrapf(err, "decodeObjectStreams: problem dereferencing object stream %d", objectNumber)
        }

        // Save decoded stream content to xRefTable.
        if err = saveDecodedStreamContent(ctx, &sd, objectNumber, *entry.Generation, true); err != nil {
            log.Read.Printf("obj %d: %s", objectNumber, err)
            return err
        }

```



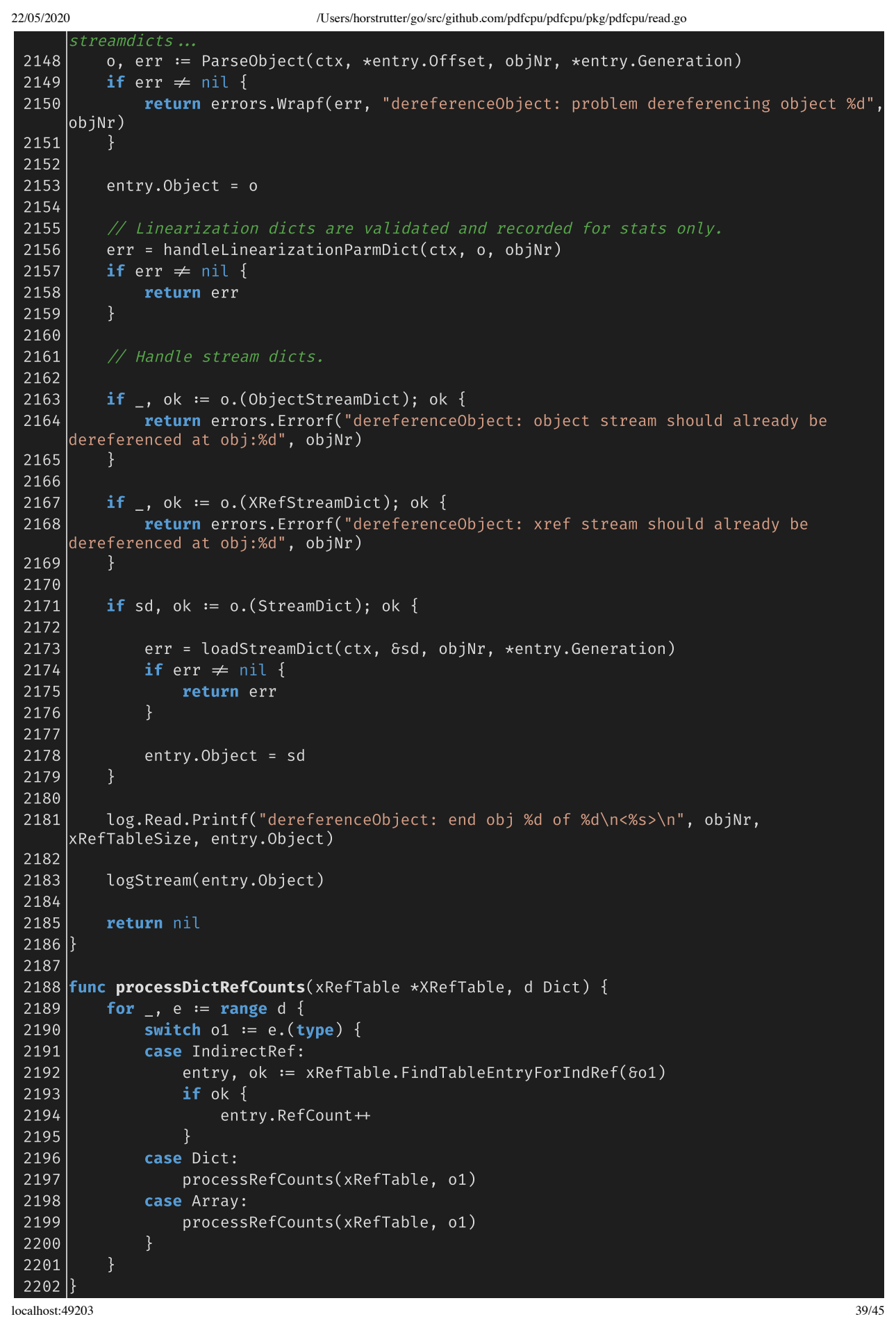
```go
        // Ensure decoded objectArray for object stream dicts.
        if !sd.IsObjStm() {
            return errors.New("pdfcpu: decodeObjectStreams: corrupt object stream")
        }

        // We have an object stream.
        log.Read.Printf("decodeObjectStreams: object stream #%d\n", objectNumber)

        ctx.Read.UsingObjectStreams = true

        // Create new object stream dict.
        osd, err := objectStreamDict(&sd)
        if err != nil {
            return errors.Wrapf(err, "decodeObjectStreams: problem dereferencing object stream %d", objectNumber)
        }

        log.Read.Printf("decodeObjectStreams: decoding object stream %d:\n", objectNumber)

        // Parse all objects of this object stream and save them to ObjectStreamDict.ObjArray.
        if err = parseObjectStream(osd); err != nil {
            return errors.Wrapf(err, "decodeObjectStreams: problem decoding object stream %d\n", objectNumber)
        }

        if osd.ObjArray == nil {
            return errors.Wrap(err, "decodeObjectStreams: objArray should be set!")
        }

        log.Read.Printf("decodeObjectStreams: decoded object stream %d:\n", objectNumber)

        // Save object stream dict to xRefTableEntry.
        entry.Object = *osd
    }

    log.Read.Println("decodeObjectStreams: end")

    return nil
}

func handleLinearizationParmDict(ctx *Context, obj Object, objNr int) error {

    if ctx.Read.Linearized {
        // Linearization dict already processed.
        return nil
    }

    // handle linearization parm dict.
    if d, ok := obj.(Dict); ok && d.IsLinearizationParmDict() {

        ctx.Read.Linearized = true
        ctx.LinearizationObjs[objNr] = true
        log.Read.Printf("handleLinearizationParmDict: identified linearizationObj #%d\n", objNr)

        a := d.ArrayEntry("H")

```



```go
        if a == nil {
            return errors.Errorf("handleLinearizationParmDict: corrupt linearization dict at obj:%d - missing array entry H", objNr)
        }

        if len(a) != 2 && len(a) != 4 {
            return errors.Errorf("handleLinearizationParmDict: corrupt linearization dict at obj:%d - corrupt array entry H, needs length 2 or 4", objNr)
        }

        offset, ok := a[0].(Integer)
        if !ok {
            return errors.Errorf("handleLinearizationParmDict: corrupt linearization dict at obj:%d - corrupt array entry H, needs Integer values", objNr)
        }

        offset64 := int64(offset.Value())
        ctx.OffsetPrimaryHintTable = &offset64

        if len(a) == 4 {

            offset, ok := a[2].(Integer)
            if !ok {
                return errors.Errorf("handleLinearizationParmDict: corrupt linearization dict at obj:%d - corrupt array entry H, needs Integer values", objNr)
            }

            offset64 := int64(offset.Value())
            ctx.OffsetOverflowHintTable = &offset64
        }
    }

    return nil
}

func loadStreamDict(ctx *Context, sd *StreamDict, objNr, genNr int) error {

    var err error

    // Load encoded stream content for stream dicts into xRefTable entry.
    if _, err = loadEncodedStreamContent(ctx, sd); err != nil {
        return errors.Wrapf(err, "dereferenceObject: problem dereferencing stream %d", objNr)
    }

    ctx.Read.BinaryTotalSize += *sd.StreamLength

    // Decode stream content.
    err = saveDecodedStreamContent(ctx, sd, objNr, genNr, ctx.DecodeAllStreams)

    return err
}

func updateBinaryTotalSize(ctx *Context, o Object) {

    switch o := o.(type) {

    case StreamDict:
        ctx.Read.BinaryTotalSize += *o.StreamLength

```



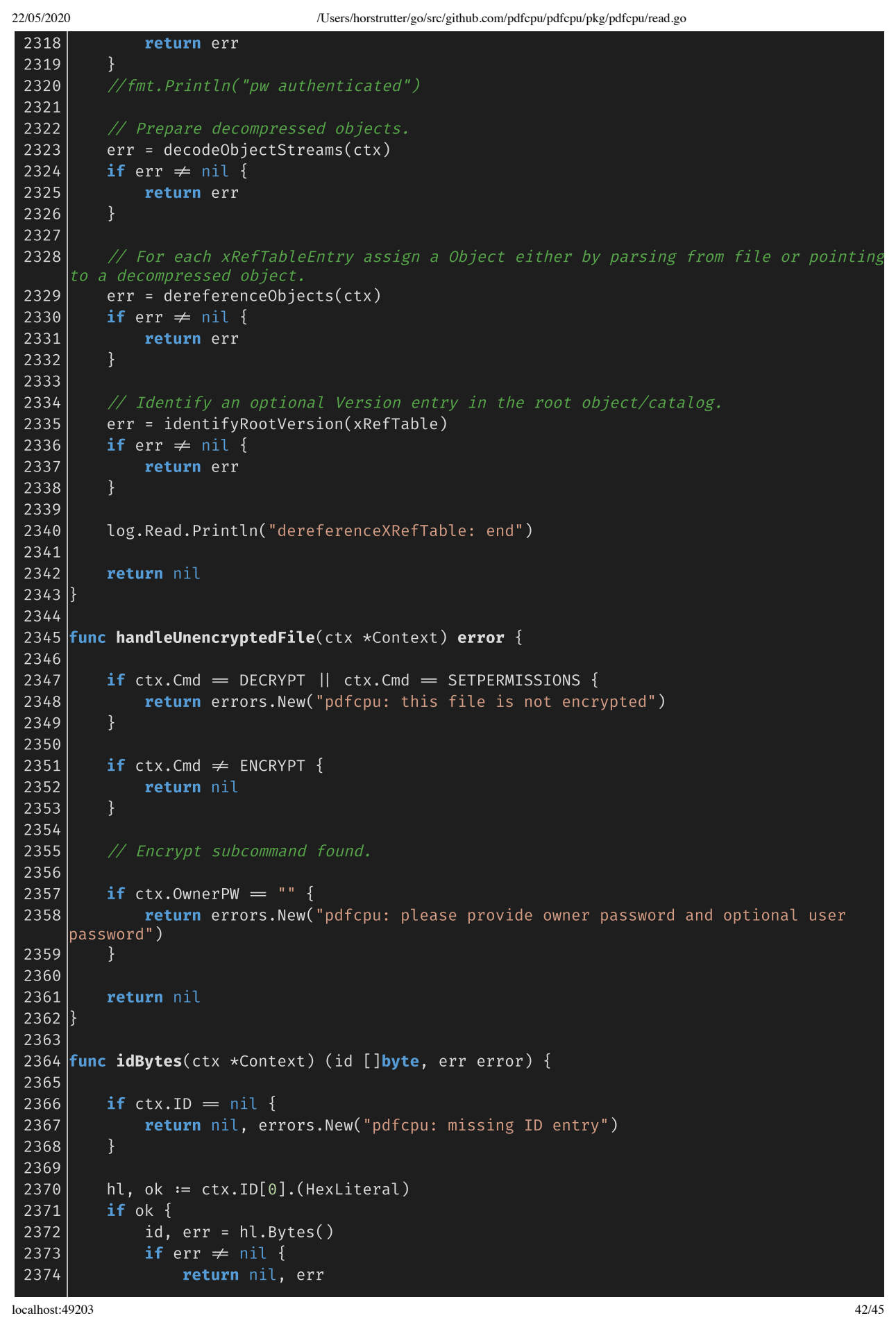
```go
    case ObjectStreamDict:
        ctx.Read.BinaryTotalSize += *o.StreamLength

    case XRefStreamDict:
        ctx.Read.BinaryTotalSize += *o.StreamLength

    }

}

func dereferenceObject(ctx *Context, objNr int) error {

    xRefTable := ctx.XRefTable
    xRefTableSize := len(xRefTable.Table)

    log.Read.Printf("dereferenceObject: begin, dereferencing object %d\n", objNr)

    entry := xRefTable.Table[objNr]

    if entry.Free {
        log.Read.Printf("free object %d\n", objNr)
        return nil
    }

    if entry.Compressed {
        err := decompressXRefTableEntry(xRefTable, objNr, entry)
        if err != nil {
            return err
        }
        //log.Read.Printf("dereferenceObject: decompressed entry, Compressed=%v\n%s\n", entry.Compressed, entry.Object)
        return nil
    }

    // entry is in use.
    log.Read.Printf("in use object %d\n", objNr)

    if entry.Offset == nil || *entry.Offset == 0 {
        log.Read.Printf("dereferenceObject: already decompressed or used object w/o offset -> ignored")
        return nil
    }

    o := entry.Object

    // Already dereferenced stream dict.
    if o != nil {
        logStream(entry.Object)
        updateBinaryTotalSize(ctx, o)
        log.Read.Printf("handleCachedStreamDict: using cached object %d of %d\n<%s>\n", objNr, xRefTableSize, entry.Object)
        return nil
    }

    // Dereference (load from disk into memory).

    log.Read.Printf("dereferenceObject: dereferencing object %d\n", objNr)

    // Parse object from file: anything goes dict, array, integer, float,
```



```go
streamdicts...
    o, err := ParseObject(ctx, *entry.Offset, objNr, *entry.Generation)
    if err != nil {
        return errors.Wrapf(err, "dereferenceObject: problem dereferencing object %d", objNr)
    }

    entry.Object = o

    // Linearization dicts are validated and recorded for stats only.
    err = handleLinearizationParmDict(ctx, o, objNr)
    if err != nil {
        return err
    }

    // Handle stream dicts.

    if _, ok := o.(ObjectStreamDict); ok {
        return errors.Errorf("dereferenceObject: object stream should already be dereferenced at obj:%d", objNr)
    }

    if _, ok := o.(XRefStreamDict); ok {
        return errors.Errorf("dereferenceObject: xref stream should already be dereferenced at obj:%d", objNr)
    }

    if sd, ok := o.(StreamDict); ok {

        err = loadStreamDict(ctx, &sd, objNr, *entry.Generation)
        if err != nil {
            return err
        }

        entry.Object = sd
    }

    log.Read.Printf("dereferenceObject: end obj %d of %d\n<%s>\n", objNr, xRefTableSize, entry.Object)

    logStream(entry.Object)

    return nil
}

func processDictRefCounts(xRefTable *XRefTable, d Dict) {
    for _, e := range d {
        switch o1 := e.(type) {
        case IndirectRef:
            entry, ok := xRefTable.FindTableEntryForIndRef(&o1)
            if ok {
                entry.RefCount++
            }
        case Dict:
            processRefCounts(xRefTable, o1)
        case Array:
            processRefCounts(xRefTable, o1)
        }
    }
}
```

```go

func processArrayRefCounts(xRefTable *XRefTable, a Array) {
    for _, e := range a {
        switch o1 := e.(type) {
        case IndirectRef:
            entry, ok := xRefTable.FindTableEntryForIndRef(&o1)
            if ok {
                entry.RefCount++
            }
        case Dict:
            processRefCounts(xRefTable, o1)
        case Array:
            processRefCounts(xRefTable, o1)
        }
    }
}

func processRefCounts(xRefTable *XRefTable, o Object) {

    switch o := o.(type) {
    case Dict:
        processDictRefCounts(xRefTable, o)

    case StreamDict:
        processDictRefCounts(xRefTable, o.Dict)

    case Array:
        processArrayRefCounts(xRefTable, o)
    }
}

// Dereferences all objects including compressed objects from object streams.
func dereferenceObjects(ctx *Context) error {

    log.Read.Println("dereferenceObjects: begin")

    xRefTable := ctx.XRefTable

    // Get sorted slice of object numbers.
    // TODO Skip sorting for performance gain.
    var keys []int
    for k := range xRefTable.Table {
        keys = append(keys, k)
    }
    sort.Ints(keys)

    for _, objNr := range keys {
        err := dereferenceObject(ctx, objNr)
        if err != nil {
            return err
        }
    }

    for _, objNr := range keys {
        entry := xRefTable.Table[objNr]
        if entry.Free || entry.Compressed {
            continue
        }
        processRefCounts(xRefTable, entry.Object)
```

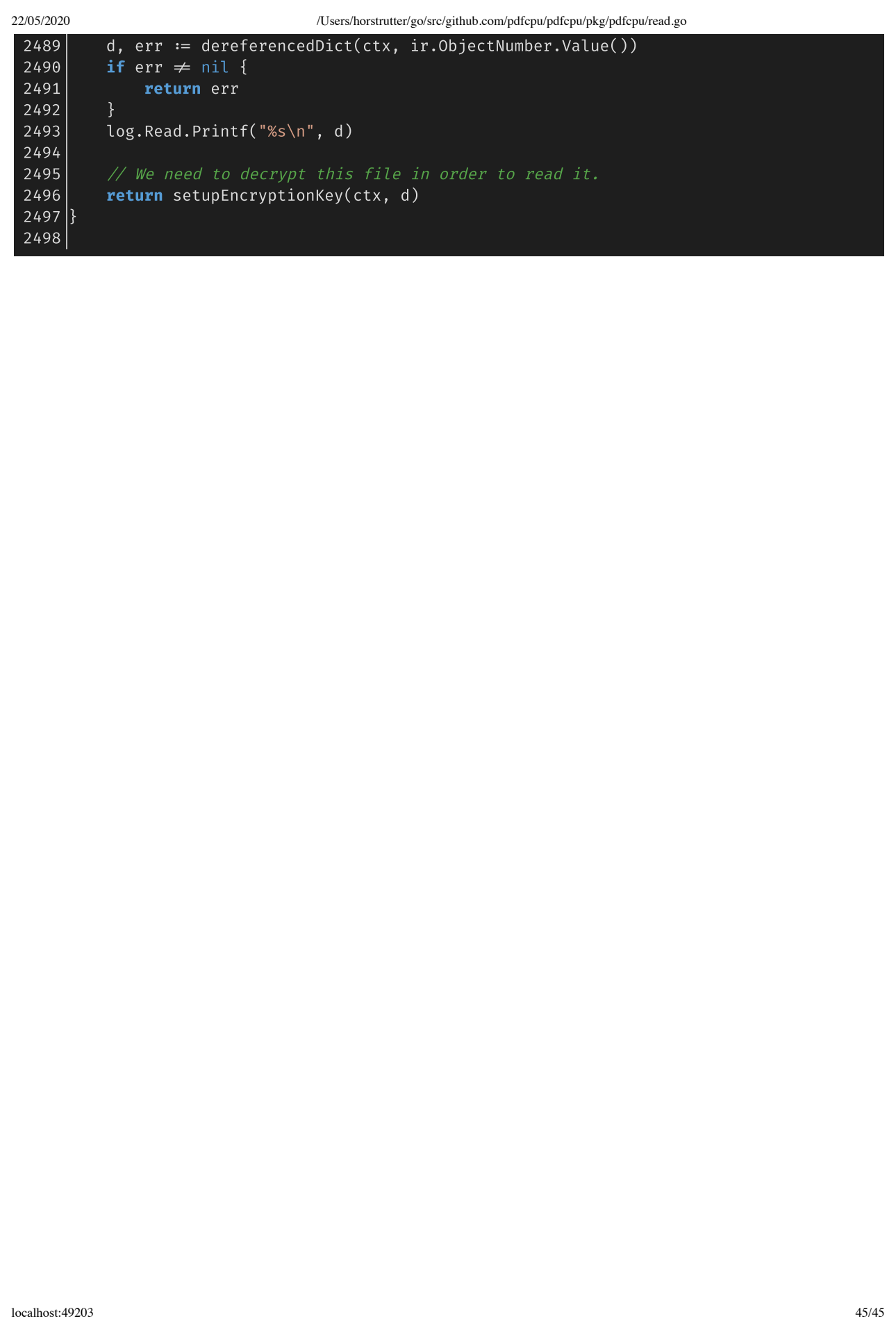
```go
    }

    log.Read.Println("dereferenceObjects: end")

    return nil
}

// Locate a possible Version entry (since V1.4) in the catalog
// and record this as rootVersion (as opposed to headerVersion).
func identifyRootVersion(xRefTable *XRefTable) error {

    log.Read.Println("identifyRootVersion: begin")

    // Try to get Version from Root.
    rootVersionStr, err := xRefTable.ParseRootVersion()
    if err != nil {
        return err
    }

    if rootVersionStr == nil {
        return nil
    }

    // Validate version and save corresponding constant to xRefTable.
    rootVersion, err := PDFVersion(*rootVersionStr)
    if err != nil {
        return errors.Wrapf(err, "identifyRootVersion: unknown PDF Root version: %s\n", *rootVersionStr)
    }

    xRefTable.RootVersion = &rootVersion

    // since V1.4 the header version may be overridden by a Version entry in the catalog.
    if *xRefTable.HeaderVersion < V14 {
        log.Info.Printf("identifyRootVersion: PDF version is %s - will ignore root version: %s\n",
            xRefTable.HeaderVersion, *rootVersionStr)
    }

    log.Read.Println("identifyRootVersion: end")

    return nil
}

// Parse all Objects including stream content from file and save to the corresponding xRefTableEntries.
// This includes processing of object streams and linearization dicts.
func dereferenceXRefTable(ctx *Context, conf *Configuration) error {

    log.Read.Println("dereferenceXRefTable: begin")

    xRefTable := ctx.XRefTable

    // Note for encrypted files:
    // Mandatory provide userpw to open & display file.
    // Access may be restricted (Decode access privileges).
    // Optionally provide ownerpw in order to gain unrestricted access.
    err := checkForEncryption(ctx)
    if err != nil {
```



```go
        return err
    }
    //fmt.Println("pw authenticated")

    // Prepare decompressed objects.
    err = decodeObjectStreams(ctx)
    if err != nil {
        return err
    }

    // For each xRefTableEntry assign a Object either by parsing from file or pointing to a decompressed object.
    err = dereferenceObjects(ctx)
    if err != nil {
        return err
    }

    // Identify an optional Version entry in the root object/catalog.
    err = identifyRootVersion(xRefTable)
    if err != nil {
        return err
    }

    log.Read.Println("dereferenceXRefTable: end")

    return nil
}

func handleUnencryptedFile(ctx *Context) error {

    if ctx.Cmd == DECRYPT || ctx.Cmd == SETPERMISSIONS {
        return errors.New("pdfcpu: this file is not encrypted")
    }

    if ctx.Cmd != ENCRYPT {
        return nil
    }

    // Encrypt subcommand found.

    if ctx.OwnerPW == "" {
        return errors.New("pdfcpu: please provide owner password and optional user password")
    }

    return nil
}

func idBytes(ctx *Context) (id []byte, err error) {

    if ctx.ID == nil {
        return nil, errors.New("pdfcpu: missing ID entry")
    }

    hl, ok := ctx.ID[0].(HexLiteral)
    if ok {
        id, err = hl.Bytes()
        if err != nil {
            return nil, err
```

```go
        }
    } else {
        sl, ok := ctx.ID[0].(StringLiteral)
        if !ok {
            return nil, errors.New("pdfcpu: ID must contain hex literals or string literals")
        }
        id, err = Unescape(sl.Value())
        if err != nil {
            return nil, err
        }
    }

    return id, nil
}

func needsOwnerAndUserPassword(cmd CommandMode) bool {

    return cmd == CHANGEOPW || cmd == CHANGEUPW || cmd == SETPERMISSIONS
}

func handlePermissions(ctx *Context) error {

    // AES256 Validate permissions
    ok, err := validatePermissions(ctx)
    if err != nil {
        return err
    }

    if !ok {
        return errors.New("pdfcpu: corrupted permissions after upw ok")
    }

    // Double check minimum permissions for pdfcpu processing.
    if !hasNeededPermissions(ctx.Cmd, ctx.E) {
        return errors.New("pdfcpu: insufficient access permissions")
    }

    return nil
}

func setupEncryptionKey(ctx *Context, d Dict) (err error) {

    ctx.E, err = supportedEncryption(ctx, d)
    if err != nil {
        return err
    }

    ctx.E.ID, err = idBytes(ctx)
    if err != nil {
        return err
    }

    var ok bool

    //fmt.Printf("opw: <%s> upw: <%s> \n", ctx.OwnerPW, ctx.UserPW)

    // Validate the owner password aka. permissions/master password.
    ok, err = validateOwnerPassword(ctx)
```

```go
    if err != nil {
        return err
    }

    // If the owner password does not match we generally move on if the user password is correct
    // unless we need to insist on a correct owner password due to the specific command in progress.
    if !ok && needsOwnerAndUserPassword(ctx.Cmd) {
        return errors.New("pdfcpu: please provide the owner password with -opw")
    }

    // Generally the owner password, which is also regarded as the master password or set permissions password
    // is sufficient for moving on. A password change is an exception since it requires both current passwords.
    if ok && !needsOwnerAndUserPassword(ctx.Cmd) {
        // AES256 Validate permissions
        ok, err = validatePermissions(ctx)
        if err != nil {
            return err
        }
        if !ok {
            return errors.New("pdfcpu: corrupted permissions after opw ok")
        }
        return nil
    }

    // Validate the user password aka. document open password.
    ok, err = validateUserPassword(ctx)
    if err != nil {
        return err
    }
    if !ok {
        return errors.New("pdfcpu: please provide the correct password")
    }

    //fmt.Printf("upw ok: %t\n", ok)

    return handlePermissions(ctx)
}

func checkForEncryption(ctx *Context) error {

    ir := ctx.Encrypt

    if ir == nil {
        // This file is not encrypted.
        return handleUnencryptedFile(ctx)
    }

    // This file is encrypted.
    log.Read.Printf("Encryption: %v\n", ir)

    if ctx.Cmd == ENCRYPT {
        // We want to encrypt this file.
        return errors.New("pdfcpu: this file is already encrypted")
    }

    // Dereference encryptDict.
```



```go
    d, err := dereferencedDict(ctx, ir.ObjectNumber.Value())
    if err != nil {
        return err
    }
    log.Read.Printf("%s\n", d)

    // We need to decrypt this file in order to read it.
    return setupEncryptionKey(ctx, d)
}

```